

MARELEC MS4/12

Static weigher and z-conveyor +
acceleration belt + grading belt

Instruction manual

(instruction manual MS4_12)

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1. Before you start...

1.1 This instruction manual

This documentation is the original instruction manual and contains important information about the purchased machine and instructions to operate it. MARELEC Food Technologies (MFT) is not responsible for damage caused by ignoring these instructions.

1.1.1 How to use this manual

Please take note of the following symbols used in this manual:



NOTE Shows a note with useful information

4

CAUTION! Potential dangerous situation for damage to machine and environment

!

DANGER! Risk for injury or death and damage to the machine

1.2 Precaution

!

DANGER! Read this manual before installing or using the machine. Improper use can cause serious damage to the user and the functions of the machine.

!

DANGER! Installation work, adjustments and repairs may only be carried out by a qualified and competent person.

!

DANGER! This machine may not be operated by persons having limited physical or mental capacities or influenced by drugs or alcohol.

1.3 Recycling



Disposal of the machine at end of life must be carried out in accordance with local environmental recycling regulations. Please verify if the protective packing crate for transport can be recycled.

2. Description of the machine

2.1 Intended use

The Marelec Grader MS4/12 is designed for automatically sorting fish with the properties and characteristics as described in chapter Description of the machine . Any other use as described is considered to be improper.

2.2 Type

- Machine: Marelec static weigher and z-conveyor + acceleration belts + grading belt
- Type: MS4/12
- Serial number: 3943
- Fabrication date: September 2012
- Made in Belgium by MFT, Redanweg 15, 8620 Nieuwpoort
- Distributed by Marelec (Chile) Ltda., Avenida Austral 1746, Jardin Oriente, Puerto Montt

2.3 Properties

2.3.1 MS4/12 at a glance...

The MS4/12 is a machine that sorts out fish based on the weight of the fish into maximum 12 classes. The machine contains of a static weighing system ❶, a z-conveyor ❷, two acceleration belts ❸ and a grading belt ❹. The machine is controlled by the Marelec Z2 computer.

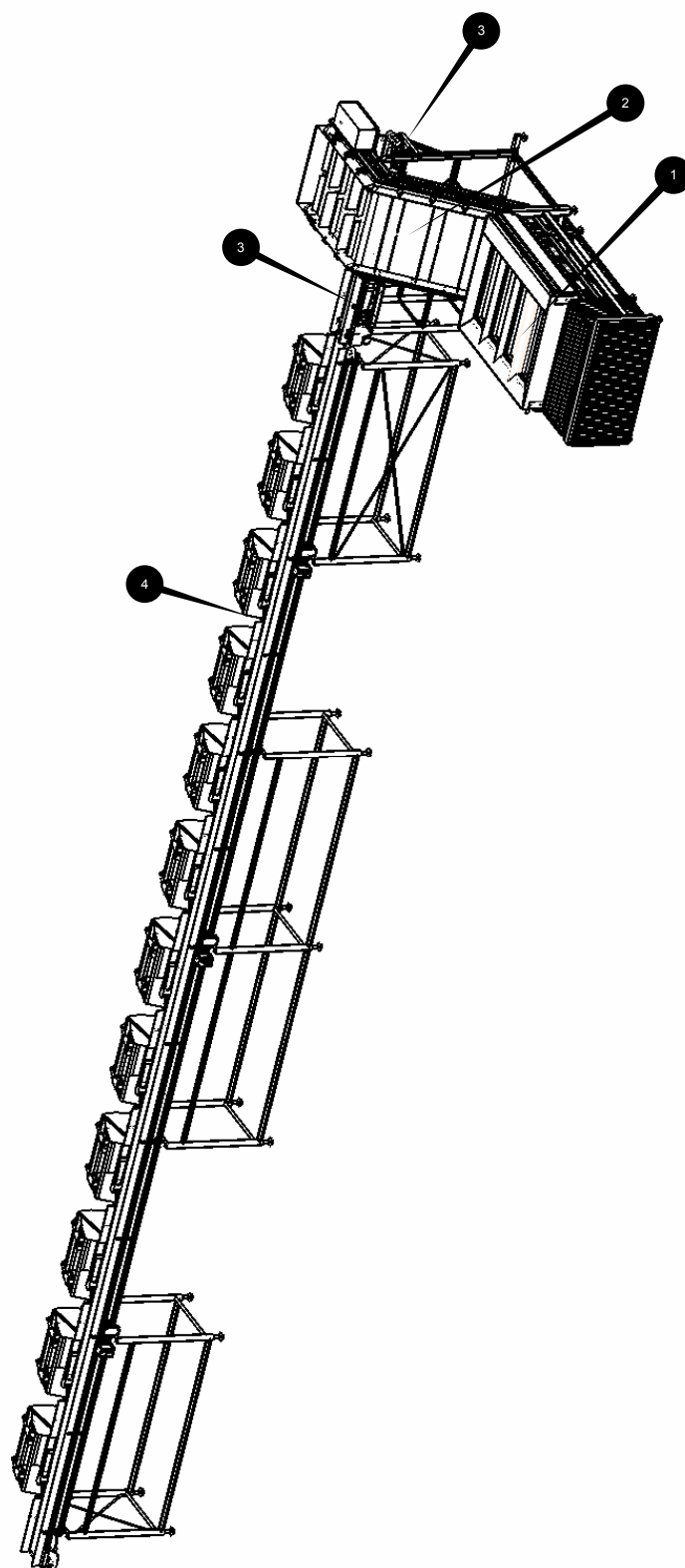


Figure 1: Marelec Grader MS4/12

2.3.2 Characteristics

- belt width:
 - z-conveyor 1200mm
 - acceleration belt 300mm
 - acceleration belt 300mm
 - grading belt 300 mm
- maximum speed: belt speed 1.4 m/sec
- maximum product dimensions: length 1100 mm, width 290mm
- outputs: 12 outputs all on right side without batching
- material: AISI 304 and food approved plastics
- weighing range: 0 – 15 kg
- precision: Range: 0 – 5kg : stdev: 20 g
5 – 15kg : stdev: 50 g

4

CAUTION!

0. Do not use products bigger than maximum product dimensions.
1. Do not use products outside weighing range.
2. Do not speed up faster than the maximum speed.

2.3.3 Outside dimensions

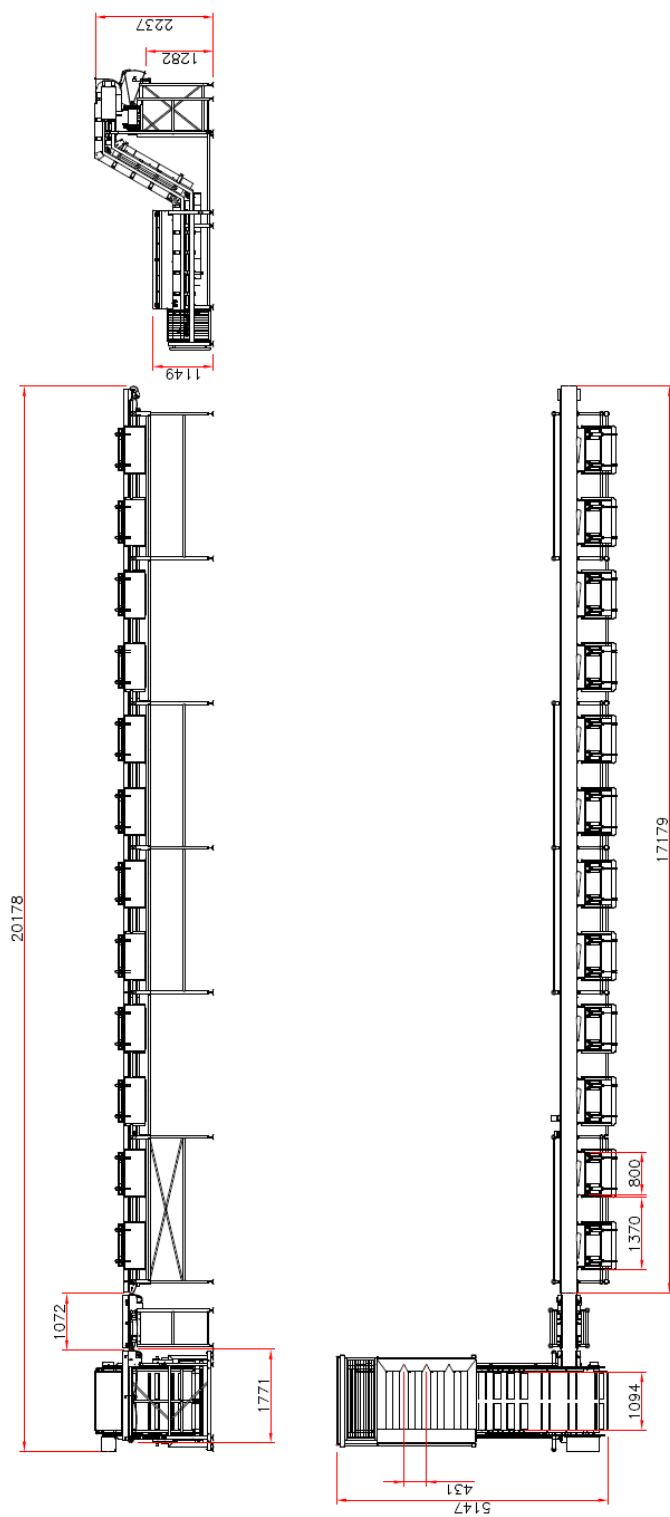


Figure 2: Outside dimensions

2.3.4 Energy

1. Electrical power (see electrical drawings).
2. Air supply of 600 norm liters at least 0,7 MPa, connection with male probe of quick acting coupler with ARO profile. Provide an compressed air dryer.

2.3.5 Safety messages

Icon	Meaning	Position on the machine	Application
	Warning for high voltage.	Access door to electrical cabinet and all components that stay under tension after turning off the machine.	All electrical parts when the machine is turned on are a potential danger for electrocution. The electrical parts that stay under tension after turning the machine off are still a potential danger for electrocution.
	Warning for rotating parts.	Near driving mechanism of belts.	People can be exposed to rotary parts of the belt mechanism. They are containing a potential risk for distorting limbs.

2.4 The environment

2.4.1 Work station

Foresee a safe work place beside the machine.

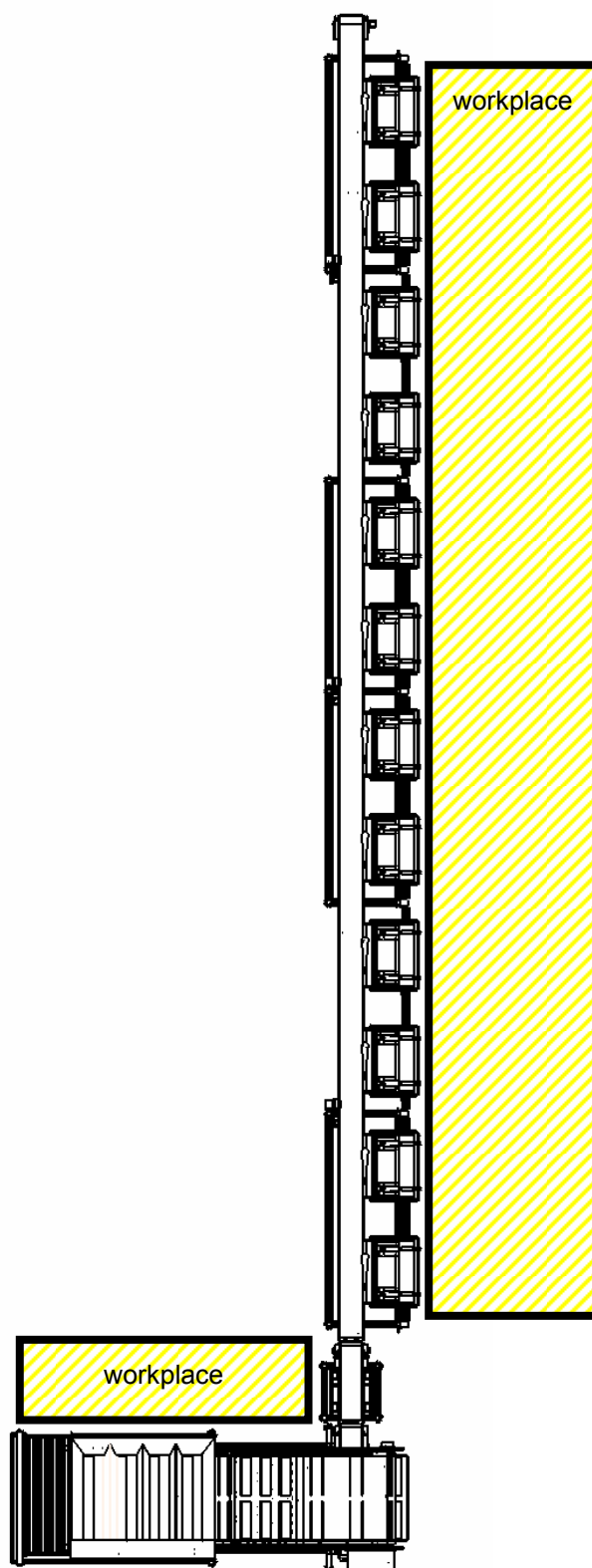


Figure 3: Work place

2.4.2 Noise emissions

The A-weighted emission sound pressure level at the work place (refer Figure 3: Work place) amounts not more than 70 dB(A).



NOTE

It's recommended to use ear protection when exposed a long time to dull noise of the machine

3. Installing the machine

3.1 Component description

3.1.1 Weighing unit

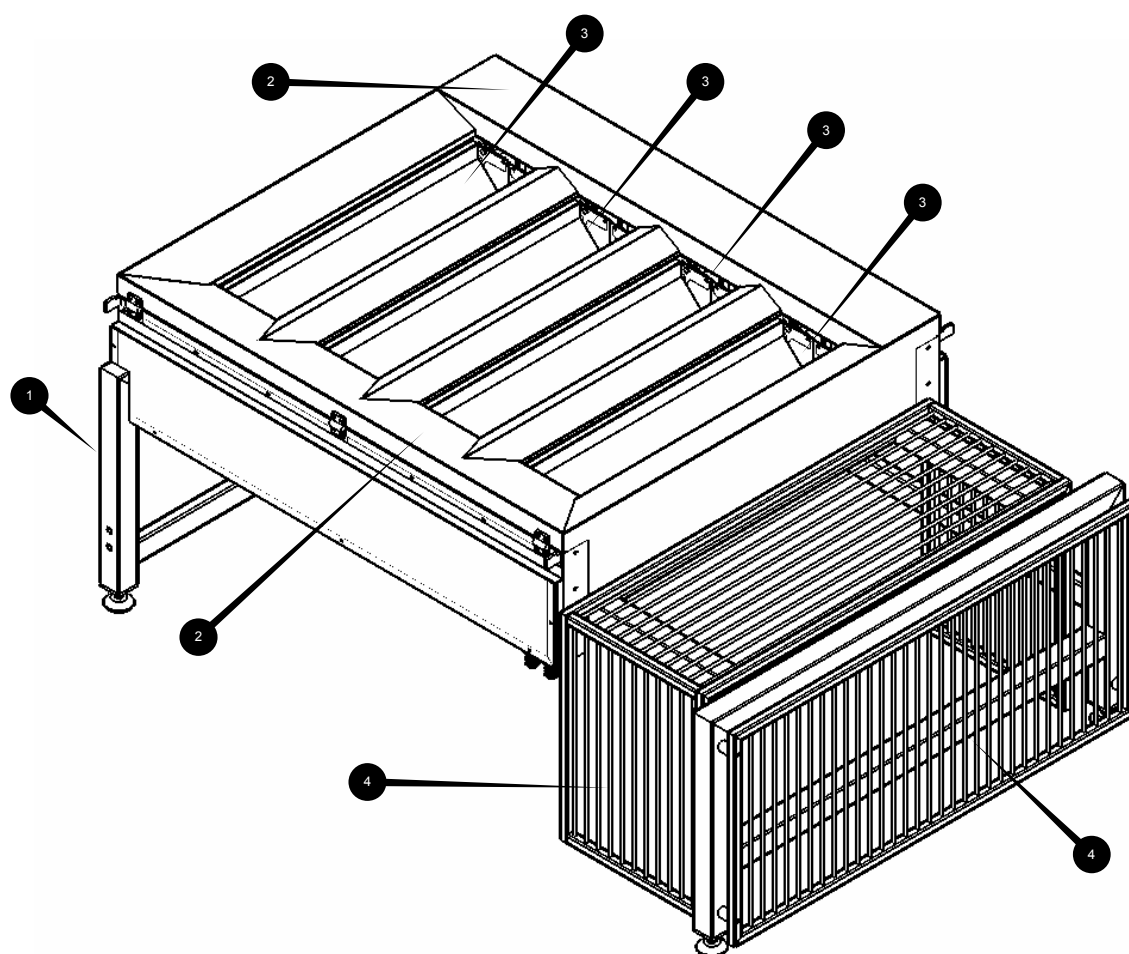


Figure 4: Weighing unit

1. Frame
2. Covers
3. Pocket
4. Safety guard

3.1.2 Z- belt

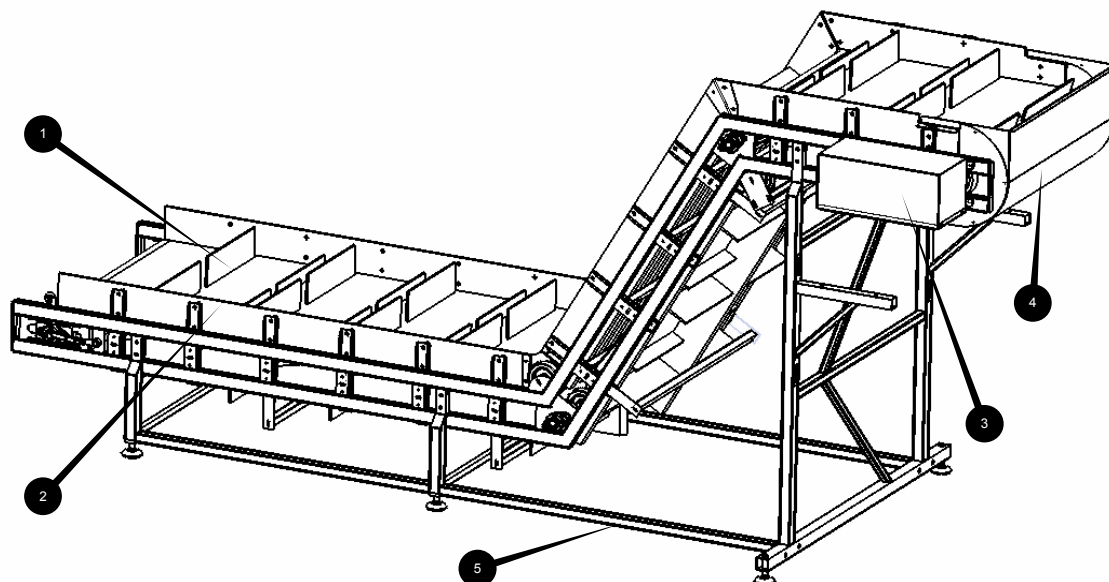
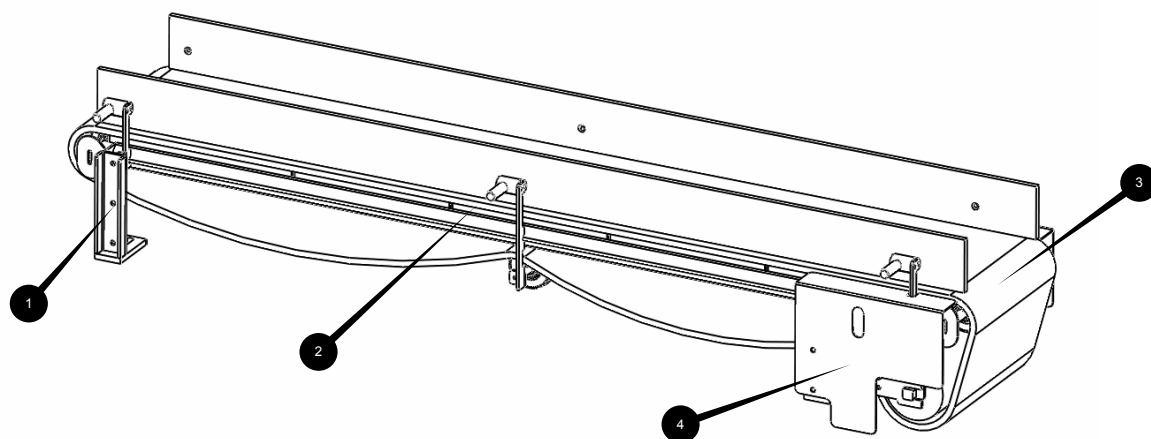


Figure 5: Z-conveyor

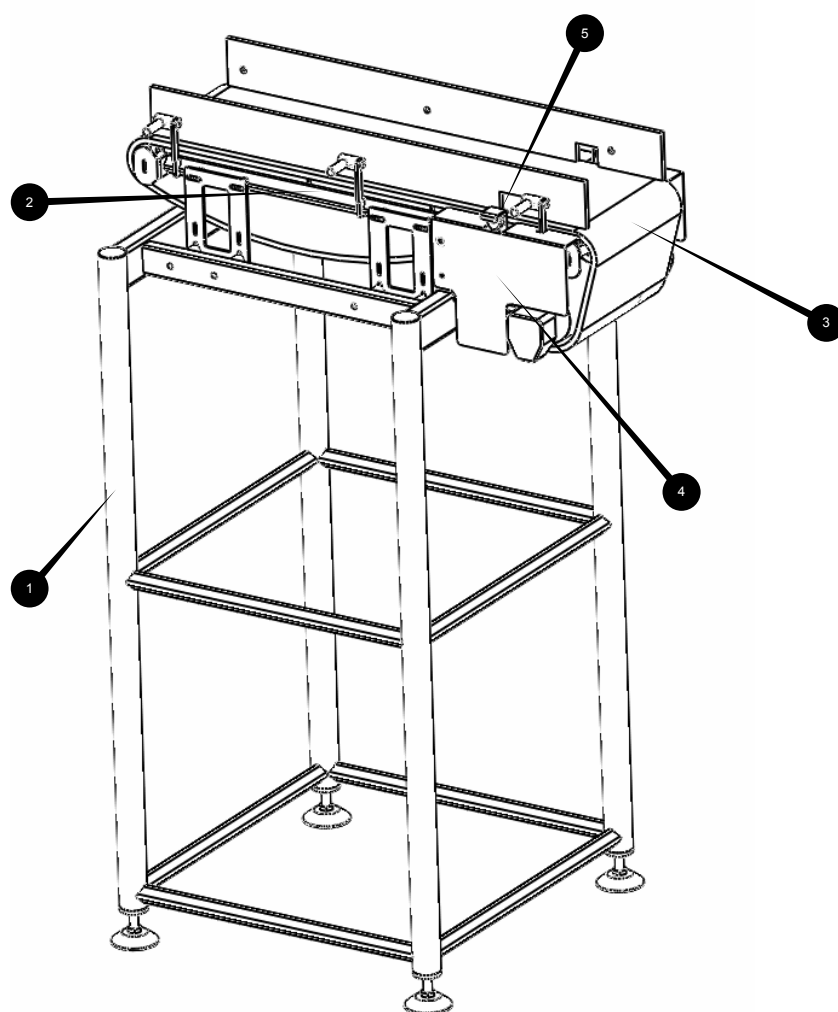
1. Belt
2. Side-guidance
3. Motorcover
4. Turn-over
5. Frame

3.1.3 First acceleration belt



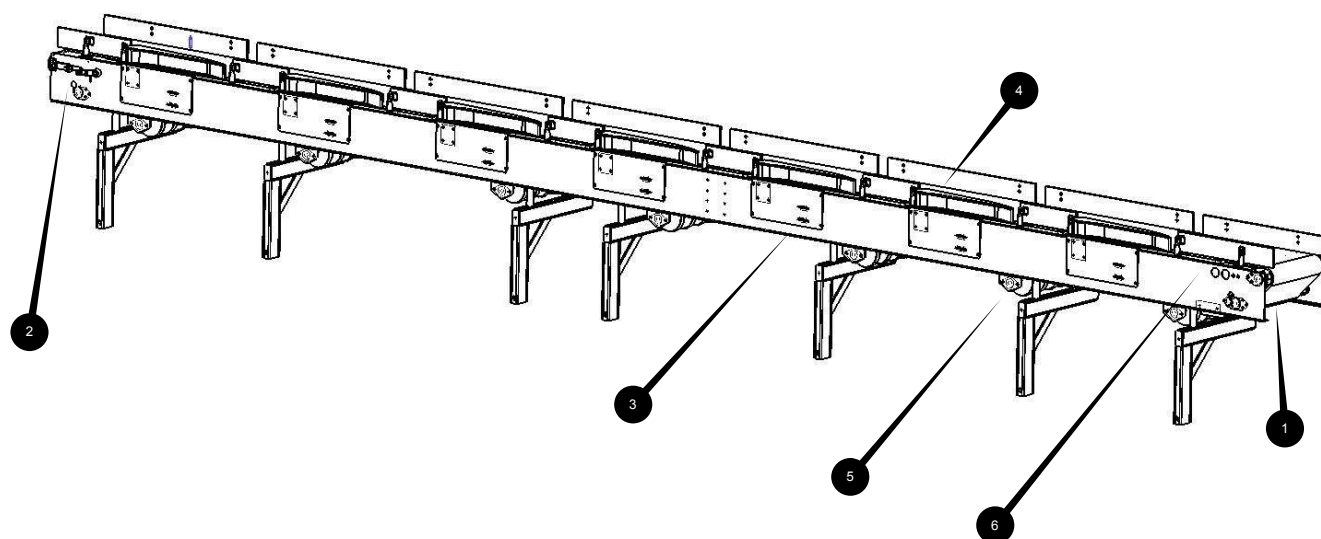
1. Bracket
2. Frame
3. Belt
4. Safety cover

3.1.4 Second acceleration belt



1. Welding construction
2. Frame
3. Belt
4. Safety cover
5. Detection eye

3.1.5 Grading belt



- 1. belt
- 0. drum motor with crank for releasing belt
- 1. flipper module
- 2. guard rail
- 3. support wheel
- 4. guiding block

Set up the machine

1. Place the second acceleration belt aligned with the grading belt and the first acceleration belt.
2. Place the weighing unit over the z-conveyor respecting the following distances:
 - Distance between two blue surfaces: 205mm

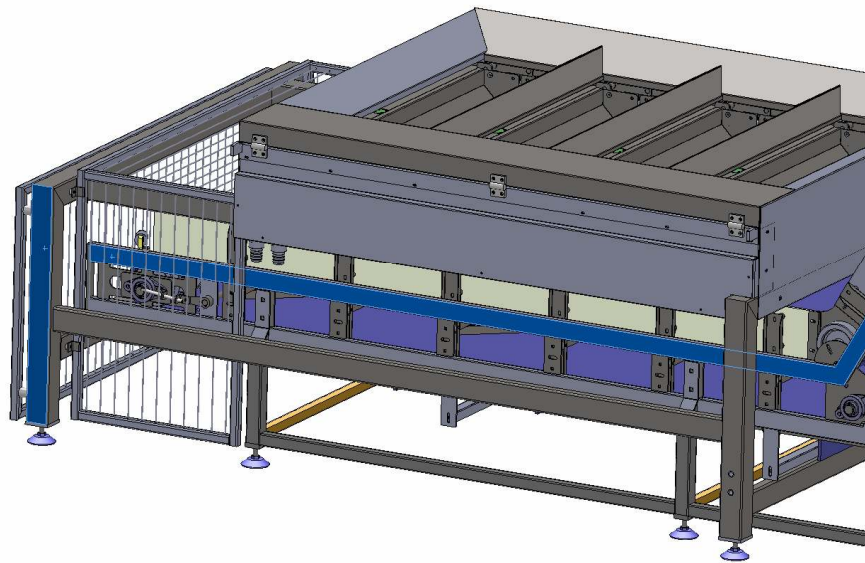


Figure 7: Positioning weighing unit

- Distance between blue surface and end part of z-conveyor: 100mm

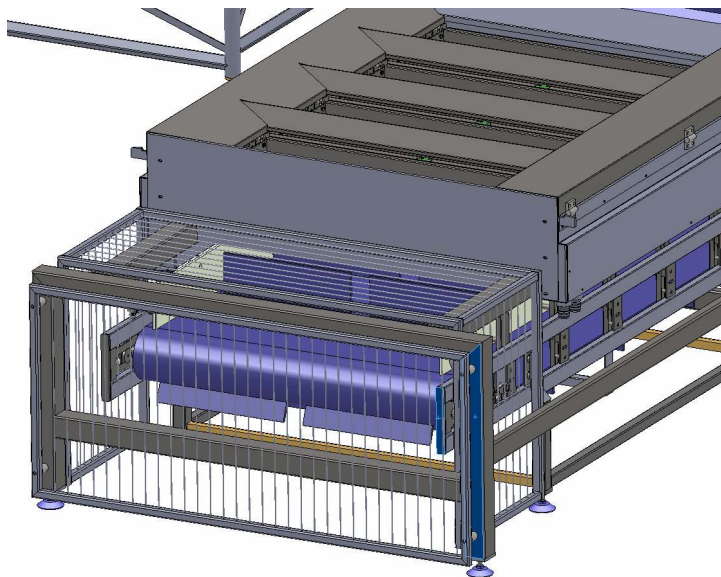


Figure 8: Positioning weighing unit

- Distance between two blue surfaces: 225mm

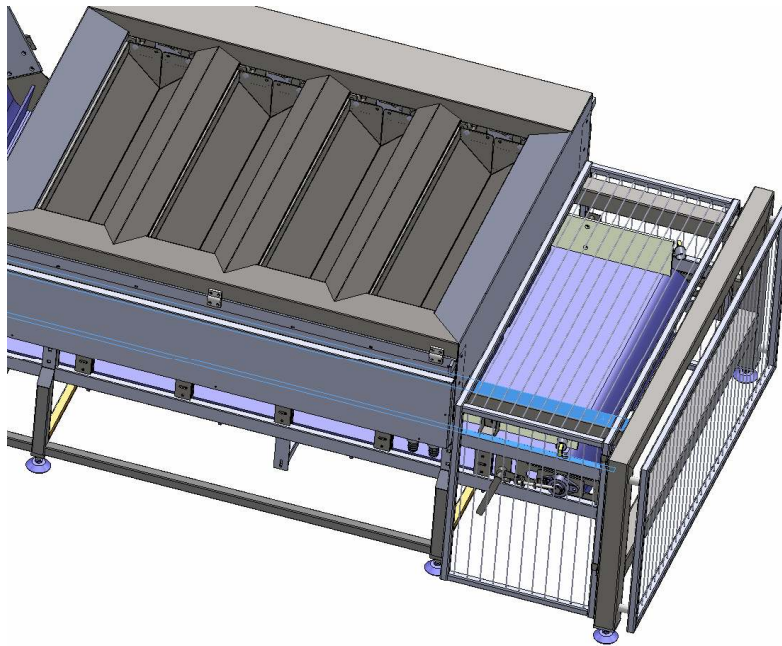


Figure 9: Positioning weighing unit

3. Insert the pockets in the weighing unit: see chapter “Installing the pockets”
4. To prevent noise emission, it's important that the belts on all machines run properly and smooth. Therefore check every passage of the belt between wearstrips and side-guidance and check the contact with the guidance-wheels. Check also the contact with the sprockets.
5. Connect electrical and pneumatic cables according electrical drawings annexed on this manual.
6. Connect power and air supply.

3.2 Installing the pockets

To install the pockets, some of the covers should be removed. In the following picture these parts are displayed blue.

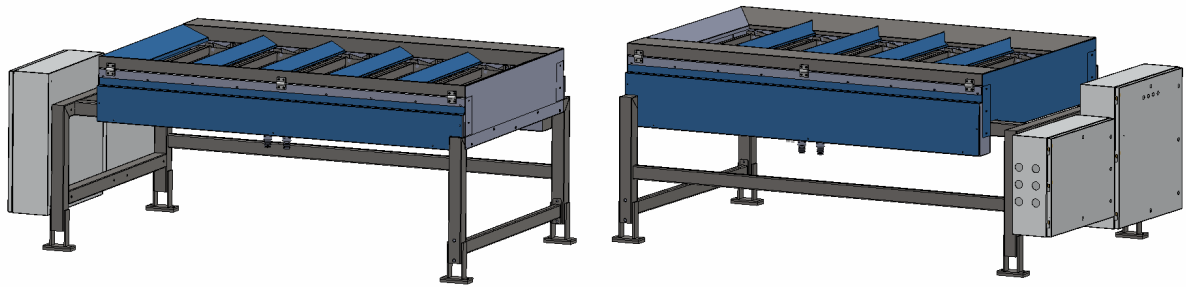


Figure 10: Removing the covers

When these parts are removed the calibrated loadcellsets can be mounted on the frame. To insert the pockets it is important to disassemble the actuators from the frame at one side of the actuator (for example the side of the hook). This ensures that the movement of the hook isn't limited anymore by the stroke of the actuator.

To insert the pockets, the hooks should be rotated, so the opening in the hooks is as vertical as possible. Now the pocket can be inserted with the wheel in the hook and it can be placed on the loadcell.

All loadcells and pockets are numbered. It is necessary to place the pocket on the correct loadcell.



Figure 11: Positioning the hooks

Once all the pockets are inserted, the actuators can be mounted on the hooks again. It is very important to check now if the wheel on the pockets is free from the hook (in the closed position of the pockets). If there is any contact between the hook and the wheel, this will influence the working of the loadcells. If the pocket is opened it is no problem when there is contact between wheel and hook.

If there is any contact between wheel and hook, the position of the actuator can be changed by loosen this 4 bolts and adjust the position of the actuator.

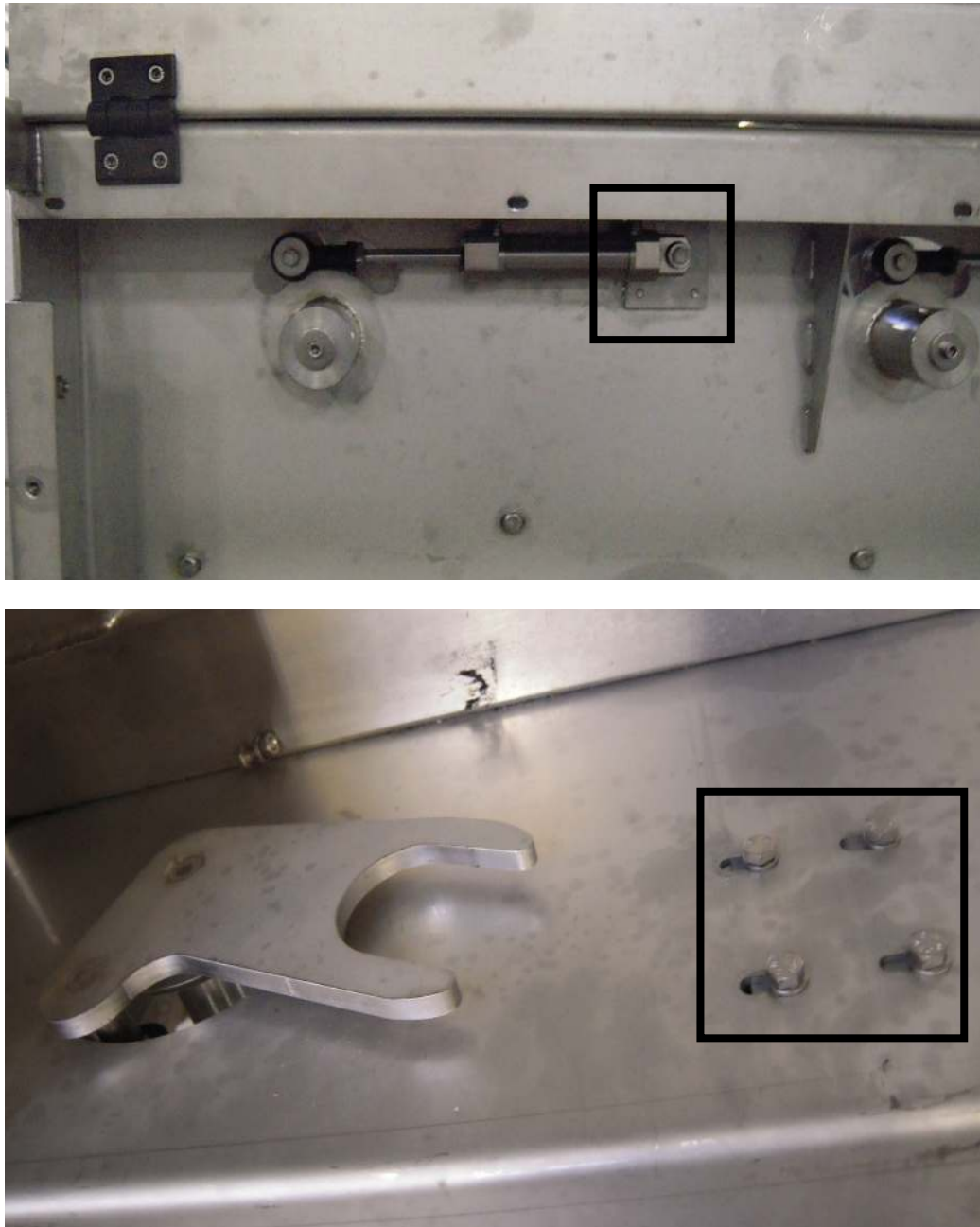


Figure 12: Adjusting the hooks

Finally the distance between hook and wheel should be like in the following picture:

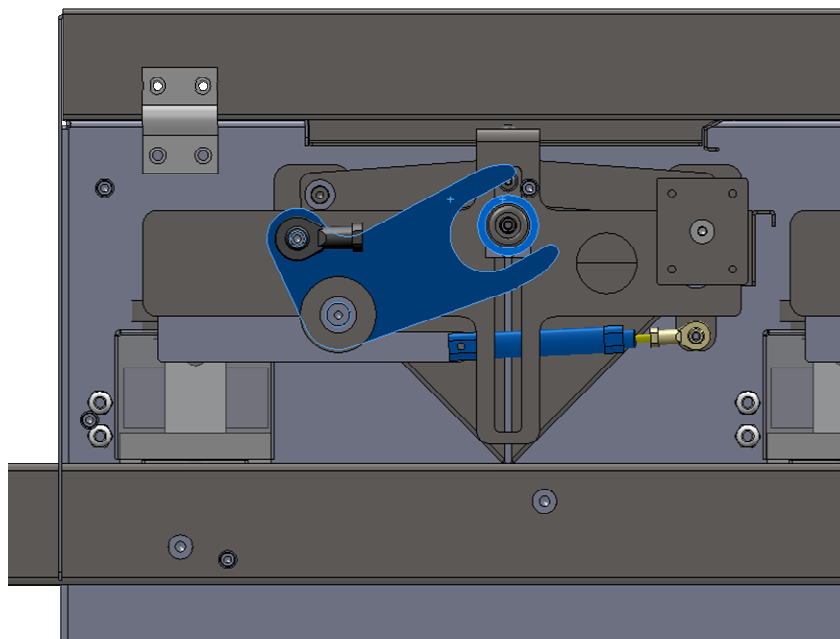


Figure 13: Correct position of the hook

Pocket shown in figure 13 may look a little bit different then the type used on your proper machine

When the pockets are mounted and the position of the actuators is adjusted in order to avoid contact between wheel and hook, all the covers that were first removed can be mounted again.

4. Operating the machine

4.1 Control devices

4.1.1 Start and stop the machine

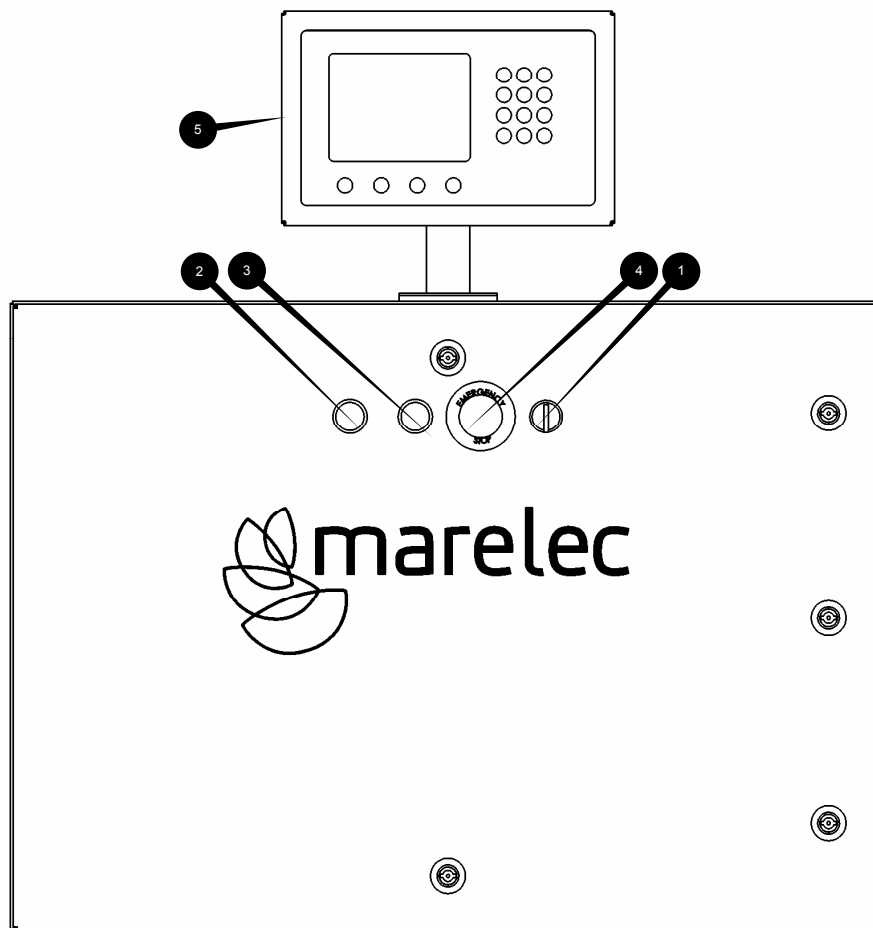


Figure 14: Electrical cabinet with control devices

1. power switch (turns power on or off)
2. start button
3. stop button
4. emergency stop
5. computer "Marelec Z2"



DANGER!

When power is turned off, several components in electric cabinet stay under tension. Danger for electrocution!

4.1.2 Marelec Z2 computer



Figure 15: Marelec Z2

1. function keys
2. numeric keyboard
3. screen



NOTE

After turning on the power, the Marelec Z2 will start up approximately 30 s. After the computer has started up, it will check all outputs. Therefore all flippers will open and close several times.



DANGER!

After the computer has started up, all flippers will open and close several times. Danger for injury and damage to machine if flippers are blocked.

For all software applications, setup and programs, see further in this manual.

4.2 Daily checkup

4.2.1 Before every use check...

1. The two detection eyes on the elevating conveyor detect a flap when it is passing by.
2. The detection eyes on the second acceleration belt detect a product when it is passing by.
3. Weight indication with empty pockets on Z2 computer between -5g and +5g.

4.2.2 After every use...

1. Clean the belt.
2. Clean detection eyes (with low pressure!).
3. Clean the entire machine (see further in this manual).

4

CAUTION!

Be careful with the waterjet when cleaning detection eyes. Although they are waterproof, they can't resist a strong waterjet

5. Cleaning

5.1 Washing

Clean the grader after every use. Conventional cleaning products can be used. However, it is absolutely necessary to rinse the machine with sweet water afterwards to avoid damage caused by chemical products.

Cleaning the machine is extremely important for the final quality of your product. This deserves attention and the cleaning staff should be well educated to perform the job correctly.

The least IP grade is IP65, however many components may have a higher protection degree. This means that high pressure cleaning on belts is permitted, but fragile components like sensors, buttons and motors should be cleaned with low pressure (less than 3 bar).

Even more important than the right cleaning product, is the correct cleaning time (in case of foam cleaning for instance) and sufficient rinsing afterwards.

When using hot cleaning, the water in your cleaning solution will evaporate quicker than on cold cleaning. Concentration of cleaning agent will rise as the water disappears, and will finally attack materials even if they are listed as compatible! Therefore instruct your staff correctly on these issues. Typical example is changes in colour of stainless steel, this is caused by overconcentration, itself caused by poor rinsing or cleaning overtime.

Cleaning shall be performed carefully in the food contact area. Any part of the machine that comes into contact with the machine, either directly or indirectly, needs proper attention during cleaning. MARELEC Food Technologies cannot take any responsibility if this is not respected.

In general, the machine has been made in an open design in favor of easy accessability and hygienic demands. Cleaning of most conveyors will not require any specific remarks, and can be done while having the belts running.

4	CAUTION!	In case of doubt about the employed cleaning product do not use any product that damages the following materials used in the machine: 1. Stainless steel AISI304 2. Polyacetal (POM) 3. Polyethylene (PE) 4. Polycarbonate (PC) 5. Polyamide (PA) Do not use cleaning products with a higher temperature than 80°C Although everything has been done to make the machine waterproof, be careful with the water jet especially in the environment of electrical cabinet, computer and pneumatic cabinet.
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5.2 Opening and closing belts

Figure shows how to open belts of the machine. Therefore take a hammer and a screwdriver to punch out the rod. (On figure 16 it is a different type of belt but the handling is the same).

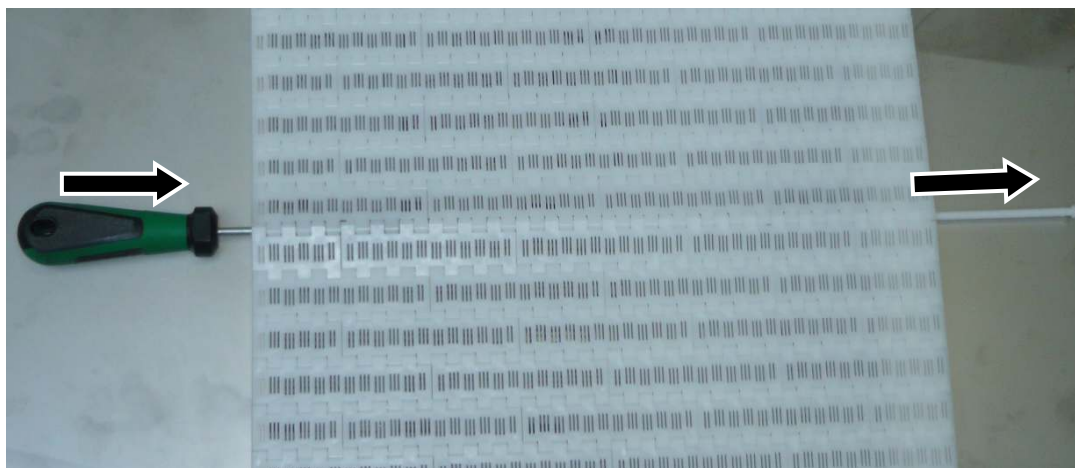


Figure 16: Opening and closing belts

6. Maintenance schedule

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CAUTION!

Ignoring maintenance instructions can cause damage to the machine, therefore follow strictly these instructions.



NOTE

For detailed instructions for all actions about maintenance, refer to Chapter Service and support

6.1 Every day

- Cleaning

Clean the entire machine after every use. See chapter Cleaning and Daily checkup for details.

- Compressed air

The compressed air used for the pockets has to be dry. Install a compressed air dryer to prevent that water comes into cylinders.



NOTE

It's not necessary to use lubrication for the cylinders as they are lubricated for life when manufactured. But once they are lubricate external, it's necessary to keep on lubricating because the original grease has flown away

6.2 Every month

- Clearance on the flippers

Check the clearance on top of the flippers. If there is a vertical clearance on the flippers, replace the bearings. If there is a horizontal clearance on the flippers then the ball bearings at the rod of the pneumatic cylinder needs to be replaced. See Figure 4: Clearance on the flippers.



NOTE

Vertical clearance is a problem if the product tends to go under the flippers.

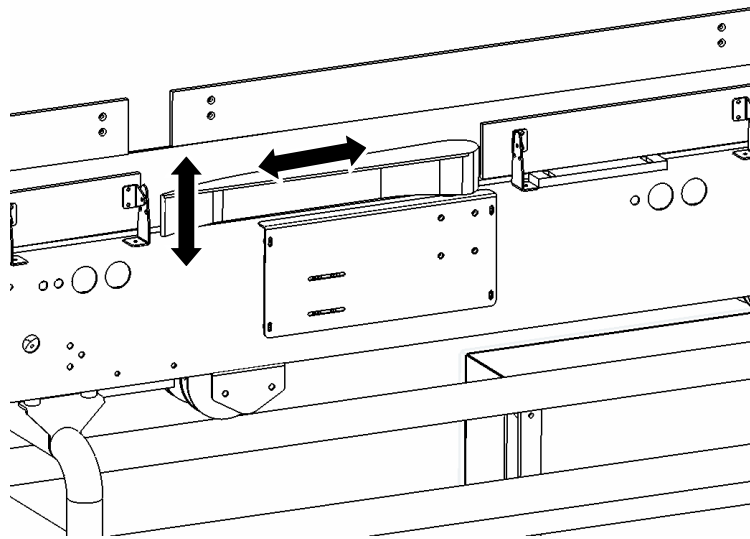


Figure 4: Clearance on the flippers

- Speed of cylinders

Check the speed of the cylinders. It has to be equal for all cylinders.

- Signal functions

Check the function of the buttons and lamps, the power switch and the emergency stop.

6.3 Every half year

- Check the modular belt.

A belt needs to be replaced when it has extended too much (too much wear on the rod of the belt). It has to be replaced also when there is too much wear on the top surface of the belt. Finally replace the belt if it's too stiff after cleaning the belt thoroughly.

- Check the loop of the modular belt.

Everywhere the belt is transferred to another part of the machine for example from the wheels to the wearstrips, the weighing plate to the transfer fingers, etc. it's necessary that the passage occurs smoothly, without vibrations and noise.

- Check the wear of the wearstrips.

7. Service and support

In the event of defects please contact your local dealer or contact the Marelec customer service:

- support@marelec.com
- tel: +32 (0)58 222 111
- fax: +32 (0)58 239 280



DANGER!

While servicing on the machine, it's important to secure all bolt – nut connections with a thread-locking compound (e.g. Loctite 243).

Risk for injury and damage to the machine because of loosened bolts caused by vibrations

7.1 Pocket composition

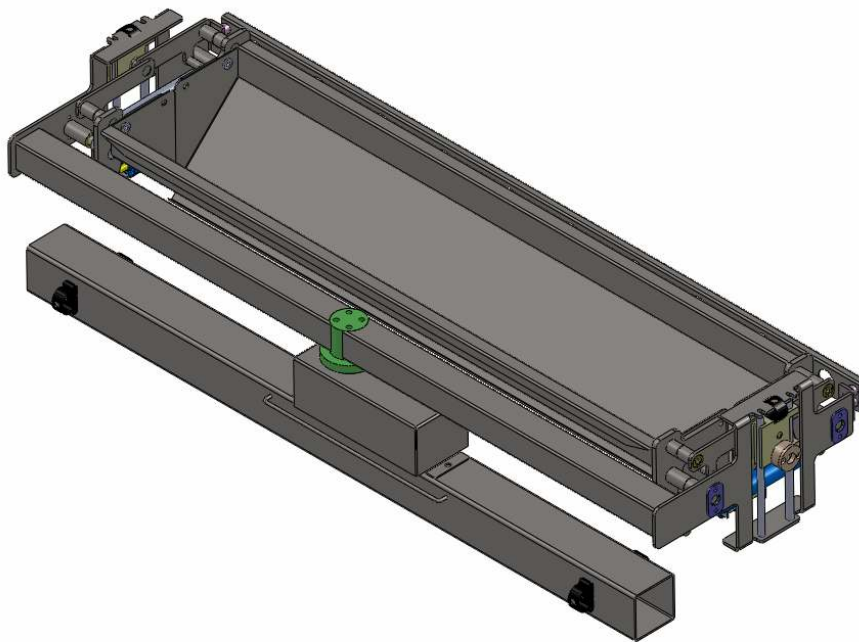


Figure 17: Pocket composition

This is how one pocket looks when everything is assembled. It consists of several important parts and the first one is the basic beam which is welded in the frame of the static weigher. On this basic beam there is a calibrated loadcell set mounted on which the basic structure of the pocket is connected.

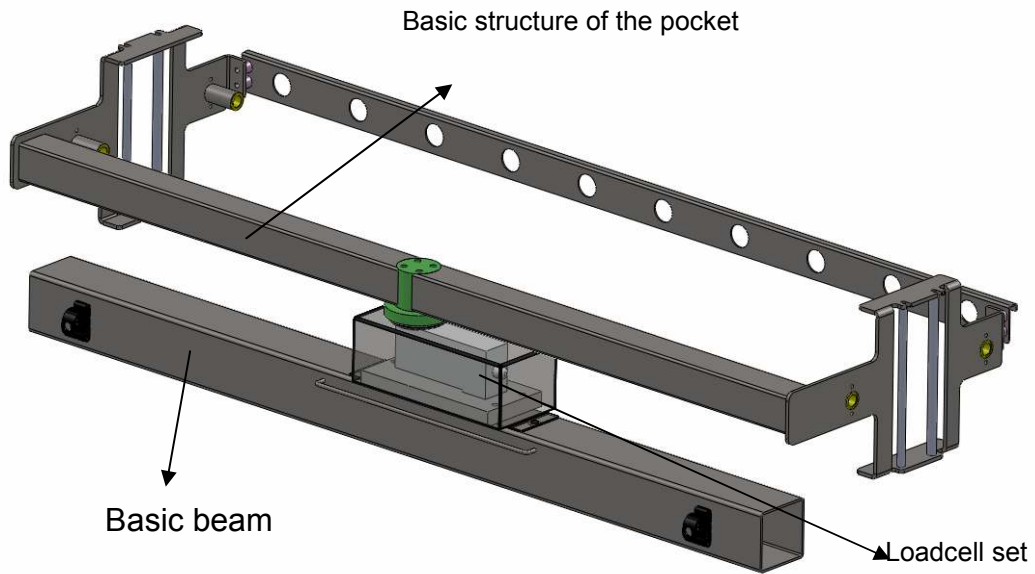


Figure 18: Basic elements of a pocket

In this basic structure of the pocket, several parts are mounted on both sides in order to form the following rod mechanism.

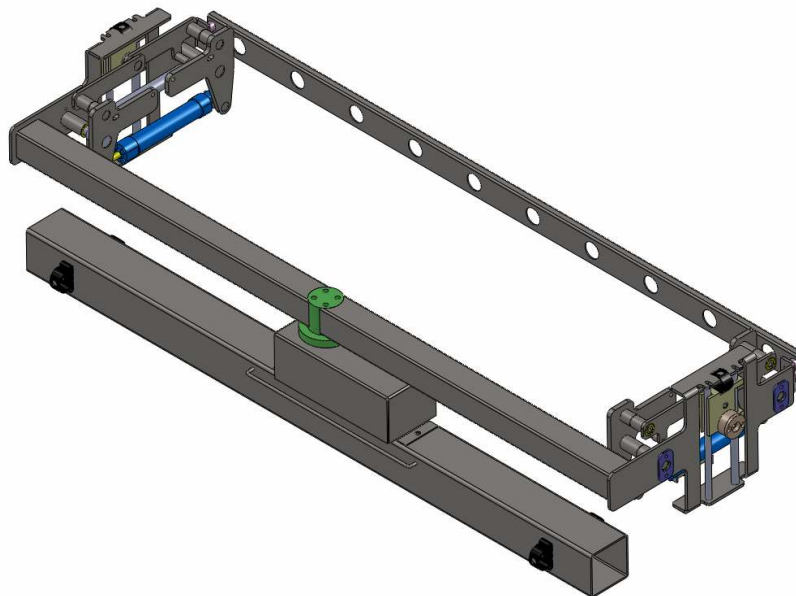


Figure 19: Rod mechanism

Once these parts are inserted, the only thing that should be done is to insert the 2 sheet metal parts in between and fix them with 2 bolts each.

Between rod e (fixed on part 3) and rod f (fixed on part 5) there is a spring. This spring makes sure that part 3 and 5 can not be fixed in a horizontal position (dead point of the mechanism). Either will parts 3 and 5 be pointing up or they will be pointing down, but they will never be horizontal.

If there is any weight in the pockets, this weight wants to make part 2 and 4 turn around rod a and b. Because of this rotation the distance between rod c and d would become smaller and also the angle between part 3 and 5 would decrease. Because part 3 and 5 are connected in part 6, part 6 would be moved upwards. The displacement of part 6 is limited with the length of the axis in part 1, so it is impossible that part 2 and 4 turn around rod a and b under influence of a weight inside the pocket.

In fact this means that how larger the weight in the pocket is, how more the pocket will be closing itself.

Part 6 can move up & downwards into part 1. This happens around 2 axis. The main axis has a diameter of 12 mm. The second axis (diameter 10 mm) has only 1 function. To make sure part 6 doesn't turn around the main axis. For this reason there is a slot into part 6, that fits around the second axis.

If the pocket must be opened, an exterior force should push part 6 downwards. Once part 3 and 5 are over their horizontal position, the weight inside the pocket and the force of the spring between rod e and f will be an extra help to open the pocket.

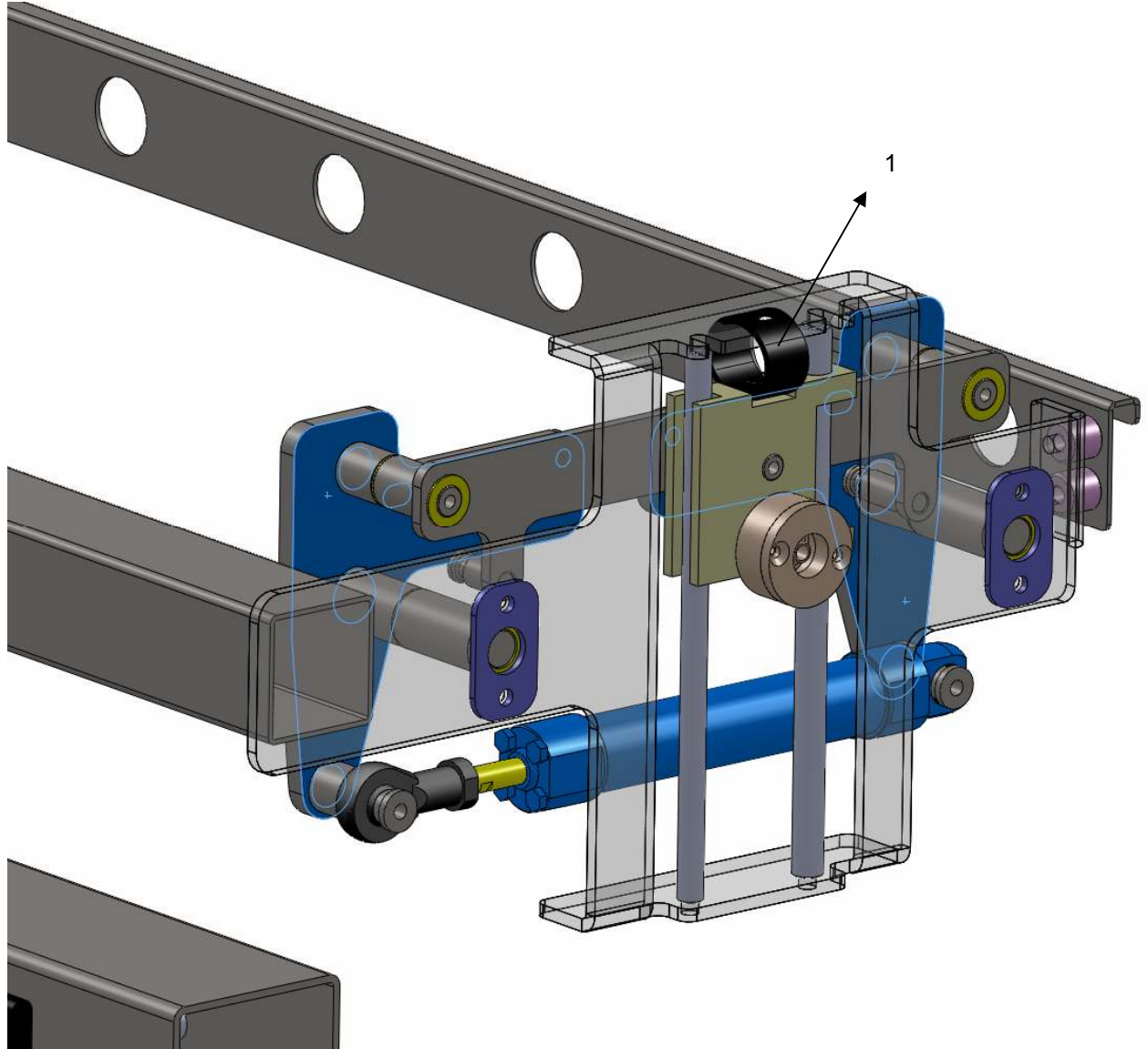


Figure 21: Adjusting the endposition of part 3 and part 5

To close the pocket again, the opposite should happen: an exterior force should move part 6 upwards until part 3 and 5 are over their horizontal position (in the opposite direction). From this moment the force of the spring makes sure that the pocket will be closed. So there is no exterior force needed for the last piece of closing the pockets.

Part 7 has a double function: it helps to eliminate the vibrations caused by the closing of the pocket and on the other side it is possible to limit the stroke of part 6 by adjusting the position of part 7. This is useful to make sure that the actuator can open the pocket with an minimal force: the weight inside the pocket helps to keep the pocket closed, but this also means that the actuator has to move this weight to open the pocket until part 3 and part 5 are in horizontal position. Once part 3 and part 5 are over this horizontal position, the weight helps

to open the pocket. For opening the pockets with a minimal force part 7 should be mounted in a way that part 3 and part 5 are just passed the horizontal position and pointing up. The ideal position is when part 3 and part 5 are as horizontal as possible, but when a force on the inside of the pocket can't open the pocket.

To be able to move part 6 in an easy way upwards or downwards, there is a little wheel mounted on this part. With a part in the shape of a hook it is possible to push the wheel up or down.

The following picture shows a vertical cut of a static weigher with this pockets. The picture shows that the blue wheel (mounted on part 6) can be moved upwards or downwards by moving the blue hook (this happens with an actuator).

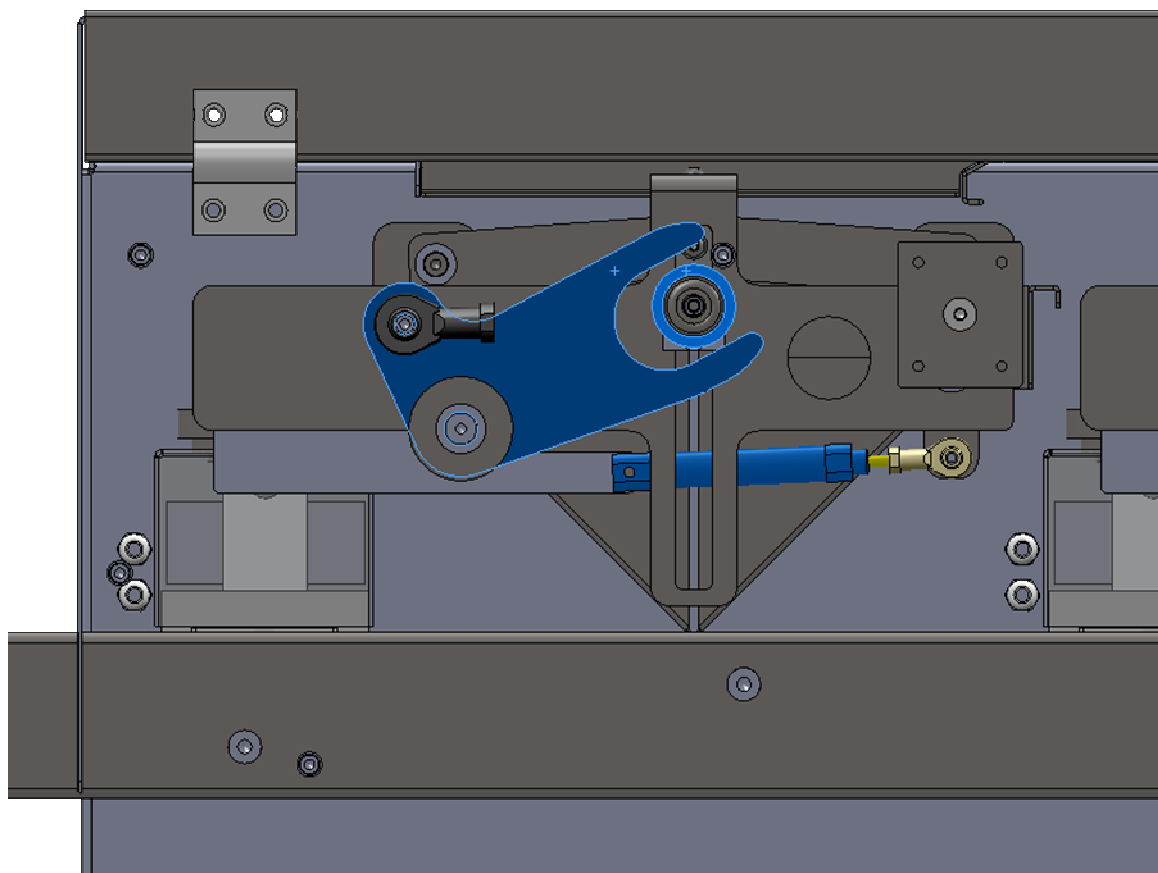


Figure 22: Distance between wheel and hook

Pocket shown in figure 22 may look different then the real model, but the principle is the same.

Because in the last piece of the displacement of part 6 the spring helps to close or open the pockets, the hook can be adjusted in a way that there is no contact between fork and wheel in the closed position.

This means that the pocket is entirely free from exterior forces during the weighing. This is an extra advantage compared with pockets where the actuators that open and close the pockets are mounted on the pocket itself. In that situation there is always some influence of the tubing of the actuators during the weighing.

Because the last part of the displacement happens under influence of the spring, this movement is not really controlled and the pockets close really fast, what can cause some vibration and noise.

To fix this there is a damper mounted between part 2 & 4 , which makes sure that opening or closing the pockets goes smooth.

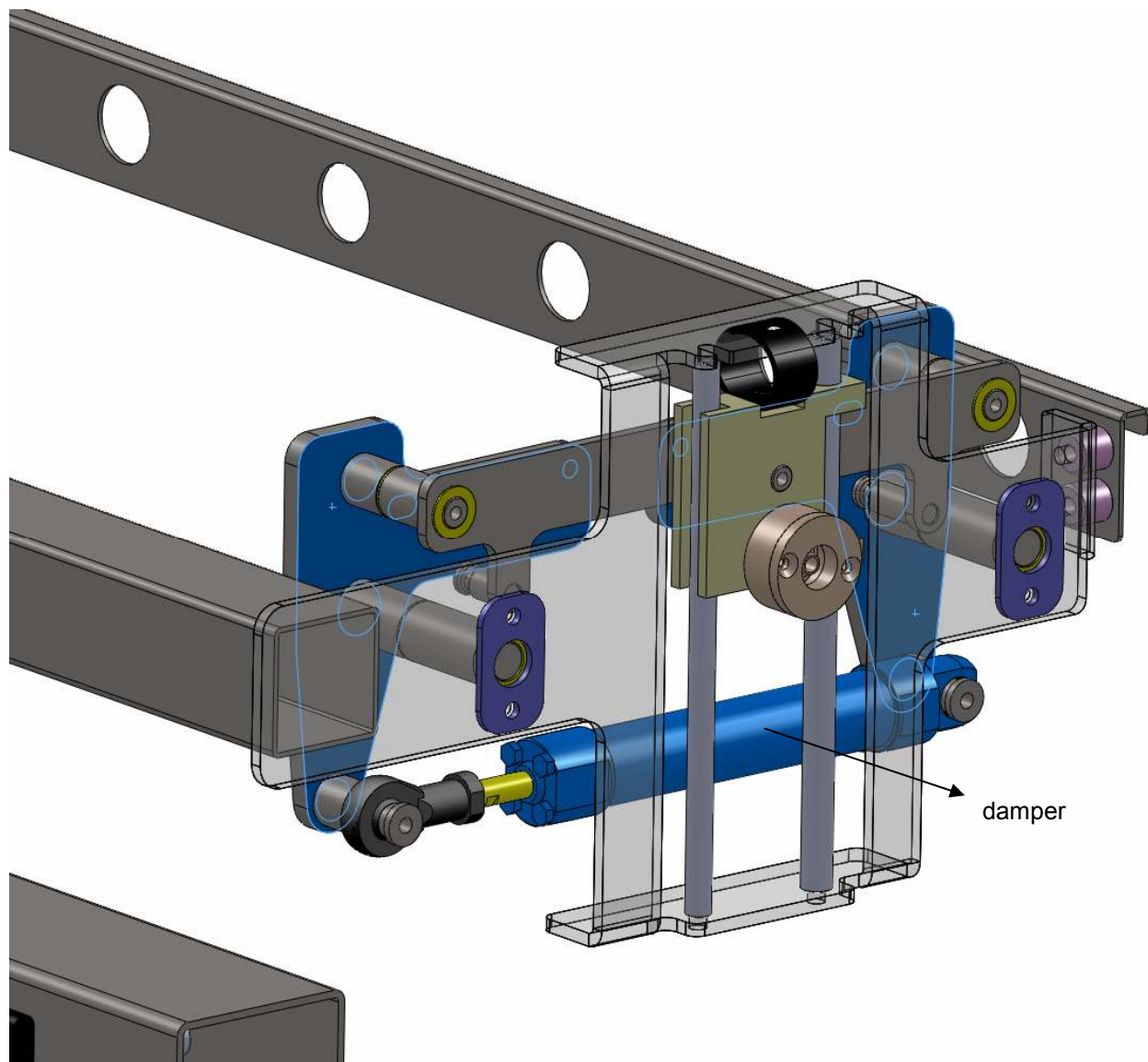
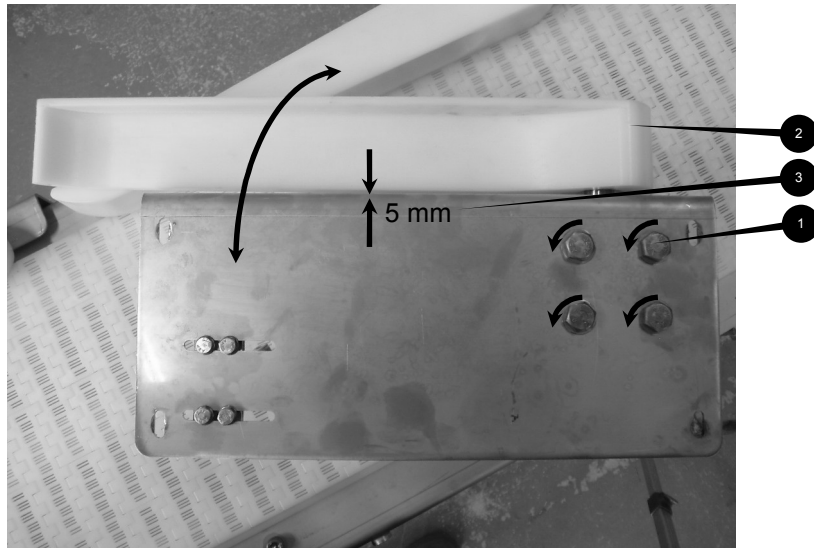


Figure 23: Damping the stroke

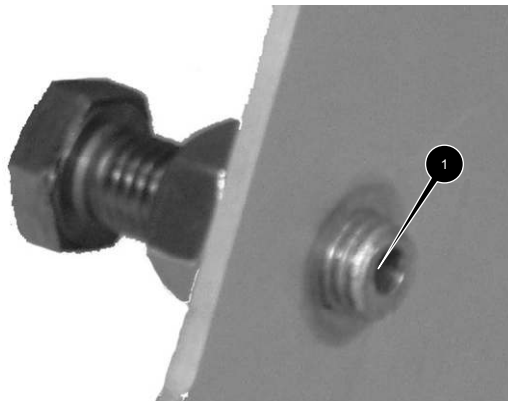
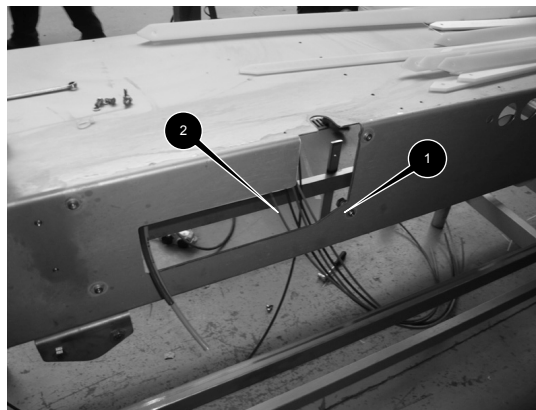
7.3 Flippers

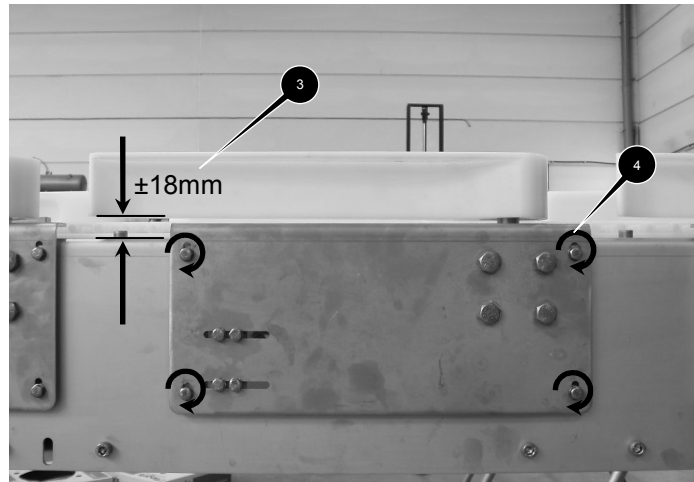
7.3.1 Install a flipper module.

6. Release 4x M10 **1** until the flipper **2** can be moved. Replace the flipper parallel about 5mm **3** above the plate and fasten 4x M10 again.

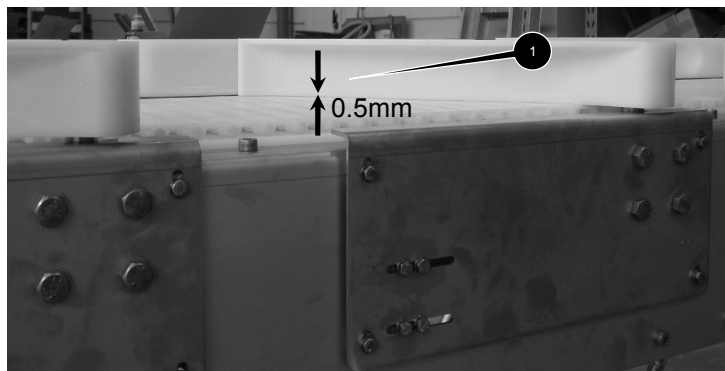


Screw M12 about 4 mm out of the frame **1**. Hang the flipper in the reserved hole **2** of the frame with retracted cylinder. Place it parallel with the top of the frame ± 18 mm **3** and tighten softly 4x M6 **4**.



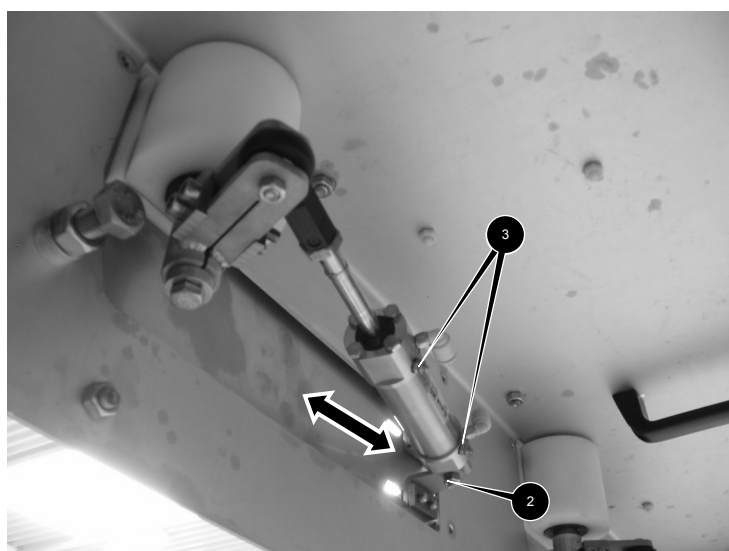


7. Now the flipper is approximately 0.5mm above the belt. If not, repeat ③④ of previous step.
8. Open the flipper. The top of the flipper must be parallel with the belt at approximately 0.5 mm ①. If not, release the nut M12 ② inside the frame and adjust M12 bolt until the top of the flipper is completely parallel at 0.5mm with the belt. After all, tighten nut M12 again.



9. Now tighten 4x M6 ④ as described in **Fout! Verwijzingsbron niet gevonden..**

10. When the cylinder is open, the flipper has to reach the other side **❶** of the belt. Therefore shift the cylinder at the attachment **❷** until it reaches the other side **❶**.

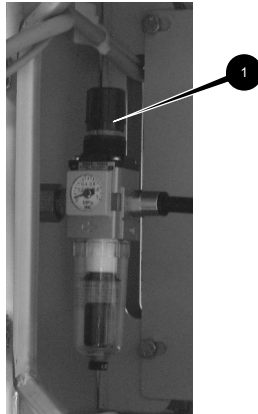


11. After connecting the pneumatic tubes, adjust the start – and end buffering of the cylinder in order that they do not smash at the end of every move **❸**.

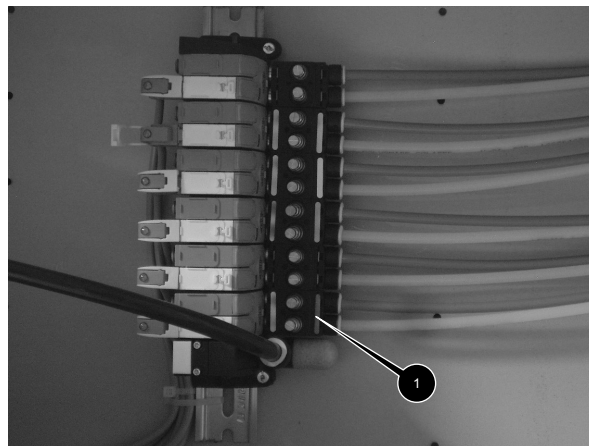
7.3.2 Adjust the speed of the cylinders

In the pneumatic cabinet the opening and closing speed of the cylinders can be adjusted. Each cylinder has its pneumatic valve with a built-in flow regulator. Follow the next steps to adjust the cylinder speed:

1. Adjust the pressure filter - regulator to at least 0.5 MPa or 5 bar ❶.



2. On each valve there is a number that corresponds with the flipper or a pocket. To adjust the speed, turn the corresponding screw of each built-in flow regulator ❶. CCW increases speed, CW reduces speed.



NOTE

1. The flippers are well adjusted if they open and close fast, but without making a vibrating noise at the open or closed position.
2. To change speed of any cylinder movement, only adjust the air flow of the exhaust tube, not the air flow that creates the force.

7.4 Belt

7.4.1 Opening and closing belts

Refer to Chapter Cleaning for details.

A belt needs to be replaced when:



NOTE

1. It has extended too much (too much wear on the rods of the belt).
2. When there is too much wear on the top surface of the belt.
3. The belt is too stiff after cleaning the belt thoroughly.

It's recommended to replace also the sprockets on drum motor when replacing the belt.

7.4.2 Acceleration belt

The belt will lengthen due to wear on the connection rods. The bearings of the outer guidance-wheels can be moved to adjust the tension.

7.4.3 Elevating belt

The belt will lengthen due to wear on the connection rods. The extra length can be compensated due adjusting the length of the lever in the following picture.

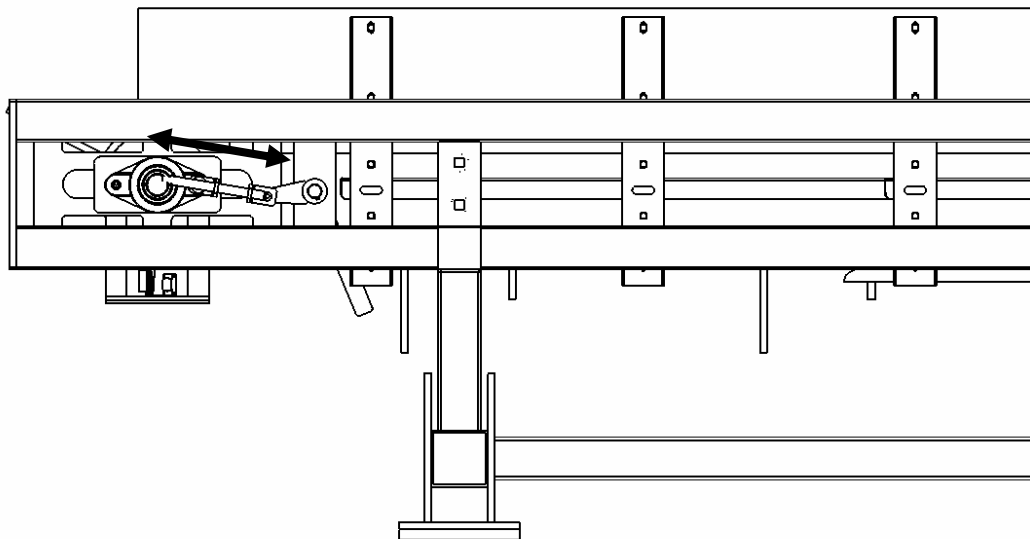


Figure 26: Adjusting the length of the lever

7.5 Drum motor

7.5.1 Mounting position

The indented arrow on the mounting shaft opposite the cable box end must point upwards (refer to Figure 5). This installation procedure ensures that the rotating internal parts are adequately submerged in oil. Any deviations from this position are allowed up to an angle of

approximately 40°. Should the angle exceed 45° the drummotor fixing shafts should be re-positioned to follow the above requirements.

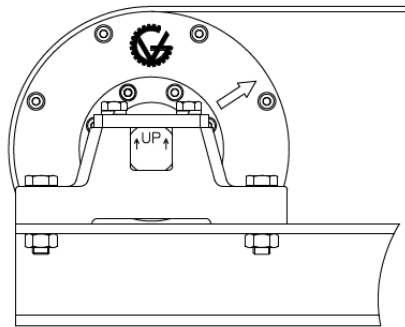


Figure 5: Mounting position

7.5.2 Changing oil

- Motor without filler plug (refer to Figure 6: Drum motor without filler plug)
4. Disconnect power supply
 5. Allow the drum motor to cool down to an acceptable and manageable temperature.
 6. Loosen the top screw ❶ from the endflange at the connection side slowly to release any air pressure within the drum, this is normal.
 7. Now loosen screw ❷. Rotate the drum motor until screw ❷ is at the 6 o'clock position and drain the oil.
 8. Rotate the motor back to its original position.
 9. Refill the drum motor with the recommended type and quantity of oil. The oil level should be just below screw ❷.
 10. Renew the copper washers ❶❷ and replace the screws.

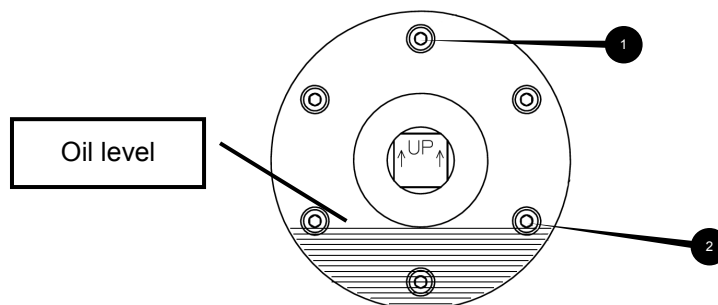


Figure 6: Drum motor without filler plug

- Motor with filler plug (refer to Figure 7: Drum motor with filler plug)
11. Disconnect power supply
 12. Allow the drum motor to cool down to an acceptable and manageable temperature.

13. Rotate the drum motor until the filler plug ❶ is at the 6 o'clock position.
14. Loosen the filler plug slowly to release any air pressure within the drum (this is normal). Remove the filler plug completely and drain the oil.
15. Rotate the drum motor until the filler plug is at the 12 o'clock position.
16. Refill the drum motor with the recommended type and quantity of oil. The radial arrow on the drum endcap where the filler plug is located must point upwards to check for the correct oil level. With the arrow in the 12 o'clock position oil should just 'weep' out the filler plug hole.
17. Replace the filler plug and renew the copper washer ❶.

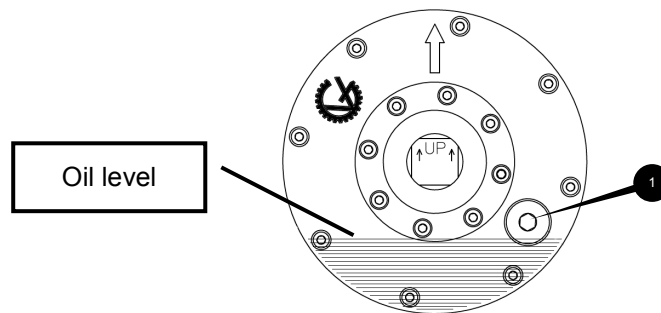


Figure 7: Drum motor with filler plug

Use oil recommended for food applications. For example:



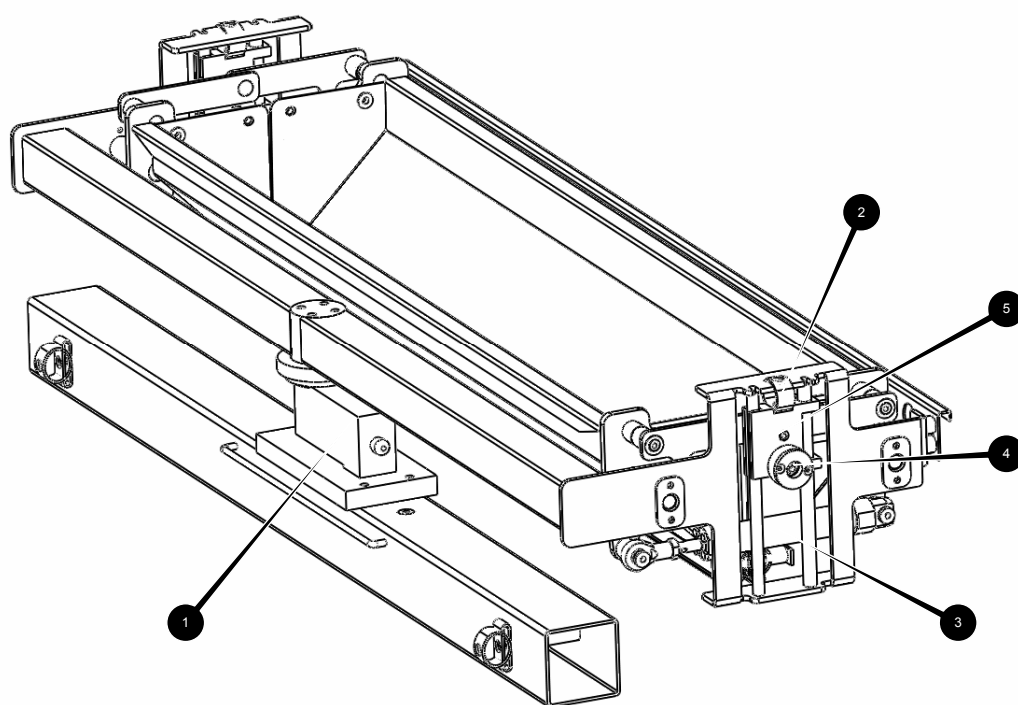
NOTE

18. Castrol Optileb GT 150
 19. Klüber UH-1-68
 20. Petro Canada Purity FG EP 100
-

8. Spare parts

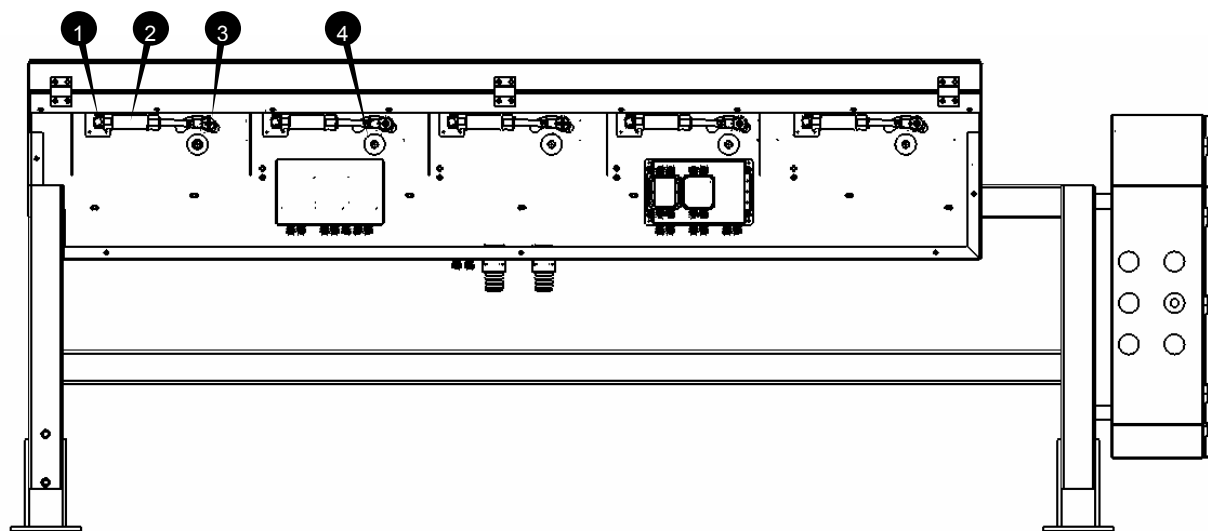
8.1 Weighing unit

- Pocket



Item	Description	Identity
1	Loadcell AK300	A404
2	Vibration damper	G916
3	Damper	G210 
4	Wheel	A1561
5	Guidance	A1562
	Spring	G503
	Igus bearings	A1564
	Rod end bearing	G16
	Bearing wheel	G1563

- Frame weighing unit

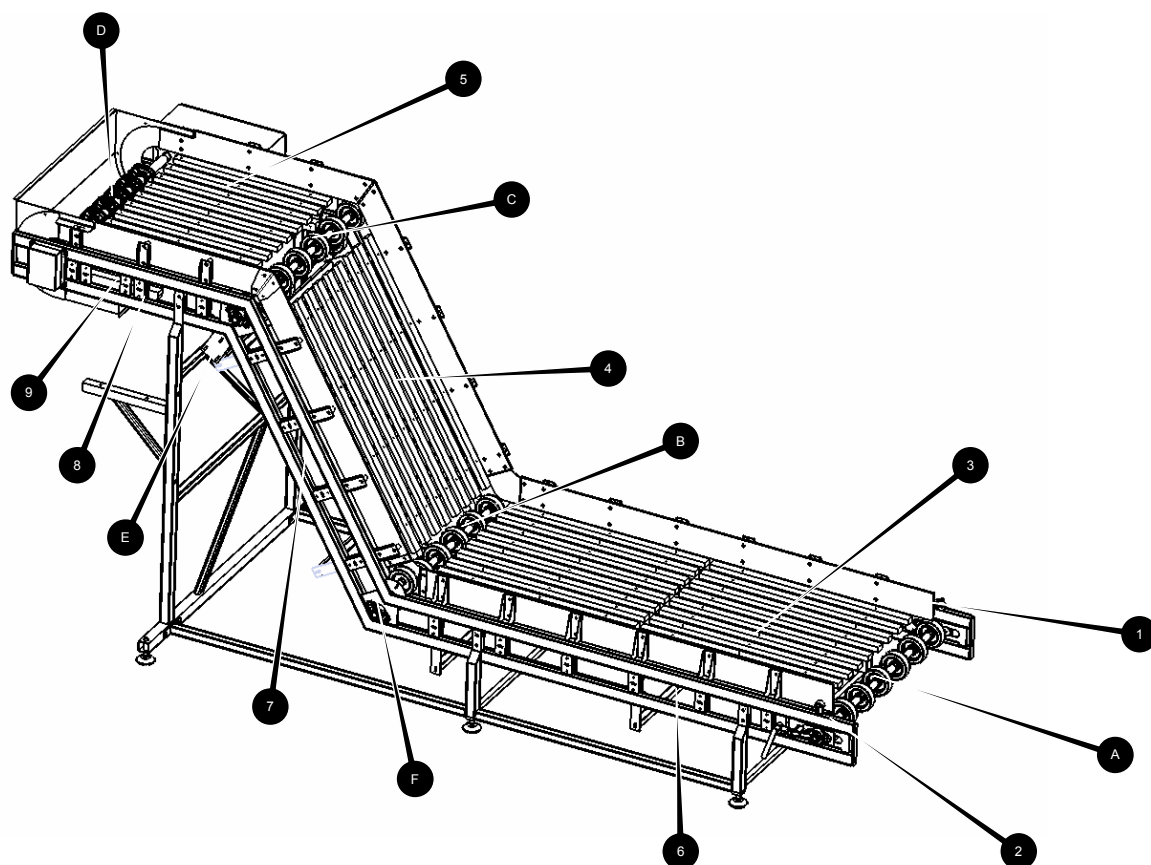


Model shown in picture above may look different to the delivered machine, but the pieces are the same

Item	Description	Identity
1	Bearing KGLM-10-SL	G531
2	Actuator CG5EA25TFSR-50	G226
3	Rod end bearing KBRM-10F	G16
4	Bearing WFM-1618-17	G532

8.2 Z-conveyor

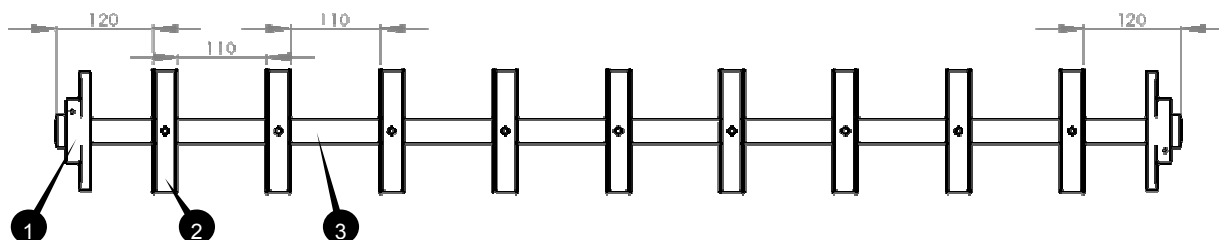
▪ General parts



Item	Description	Identity
	Belt Flush Grid 2 inch, length 2m	G1565
1	Detection eye: sender	G387
2	Detection eye: receiver	G388
3	Wearstrip 1	09/C660
4	Wearstrip 2	10/C850
5	Wearstrip 3	11/C660
6	Belt guidance 1	38/C660
7	Belt guidance 2	11/C850
8	Belt guidance 3 (long)	12/C660

9	Belt guidance 3 (short)	16/C850
A	Shaft A	
B	Shaft B	
C	Shaft C	
D	Shaft D	
E	Wheel E	
F	Wheel F	

- Shaft A



Item	Description	Identity
1	Bearing INA KLT 30 CS VA	G492
2	Wheel d=150mm	A1566
3	Shaft	01/C850

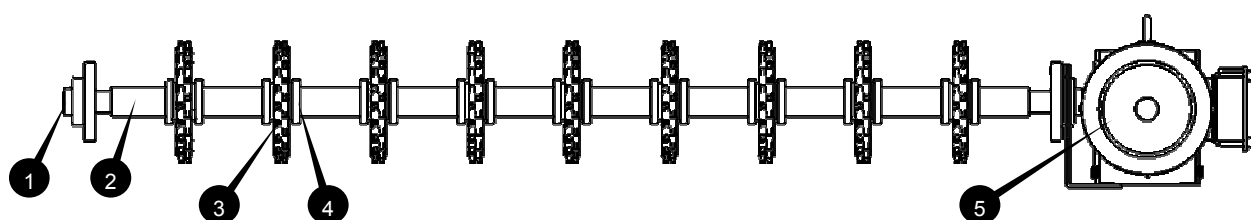
- Shaft B

Similar to shaft A

- Shaft C

Similar to shaft A & B

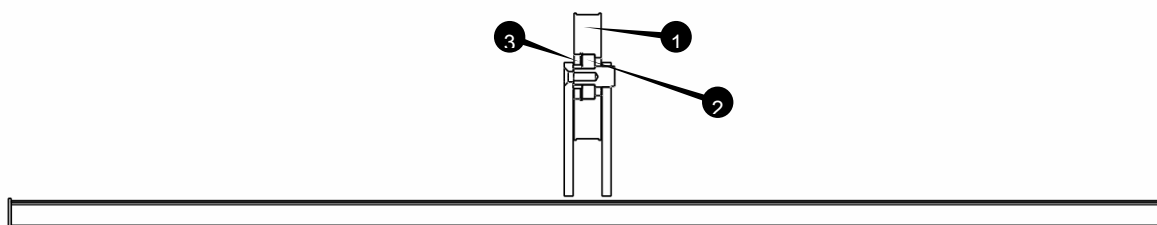
- Shaft D



Item	Description	Identity
1	Bearing F206	G509
2	Shaft	24/C660

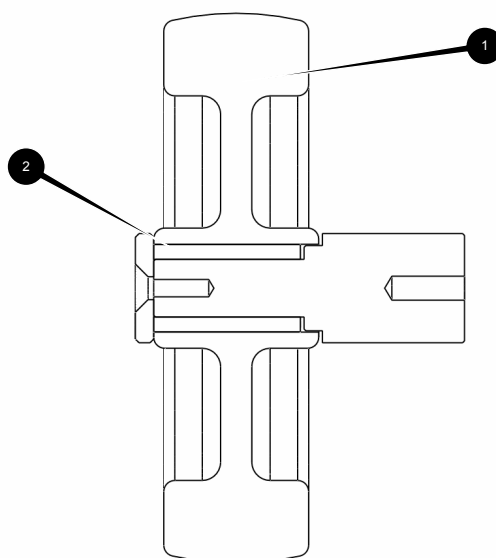
3	Sprocket M50S1060Q6	G506
3	Adaptor M50 E6040 R8	G507
4	Split shaft collar Marbett S0612 615 012	G535
5	Motor NORD 1,5kW	G502

- Wheel E



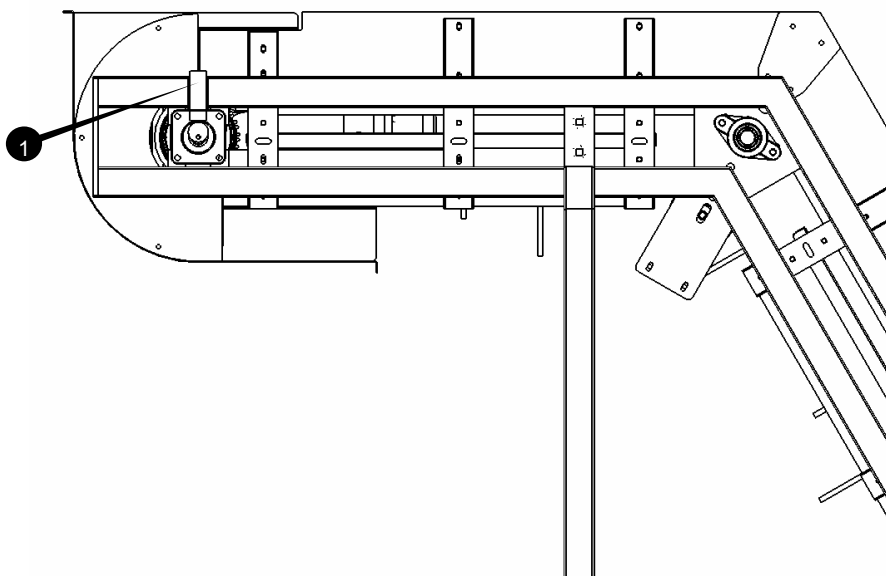
Item	Description	Identity
1	Wheel d=150mm	A912
2	Needle bearing 1624/40	G762
3	Axis	14/C770

- Wheel F



Item	Description	Identity
1	Wheel d=150mm	A912
2	Needle bearing 1624/40	G762

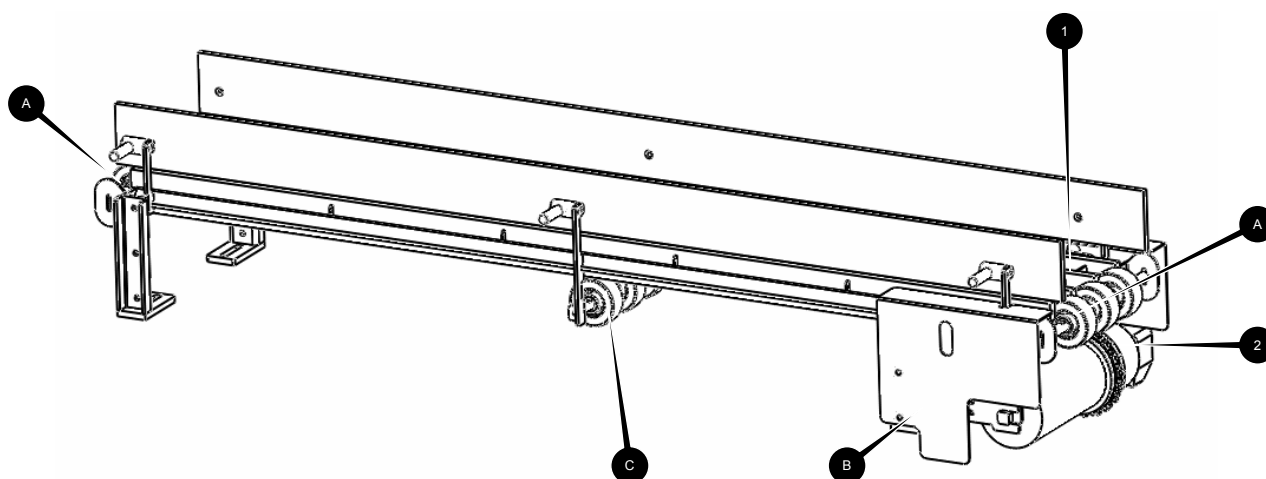
- Inductive sensor



Item	Description	Identity	Quantity
1	Inductive sensor	G485	1

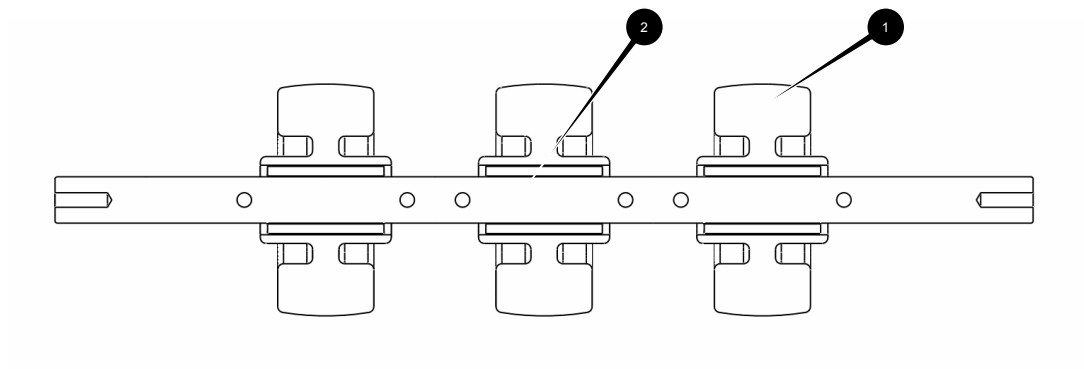
8.3 First acceleration belt

- General parts



Item	Description	Identity
	Belt M1220 High grip	G1336
	Drummotor	G1567
	Sprocket M1236	A312
1	Wearstrip	02/C851
2	Inductive sensor	G485
A	Shaft A	
B	Shaft B	
C	Shaft C	

- Shaft A



Item	Description	Identity
1	Wheel D80 mm	52/C751
2	Needle bearing 1624/40	G762

- Shaft B

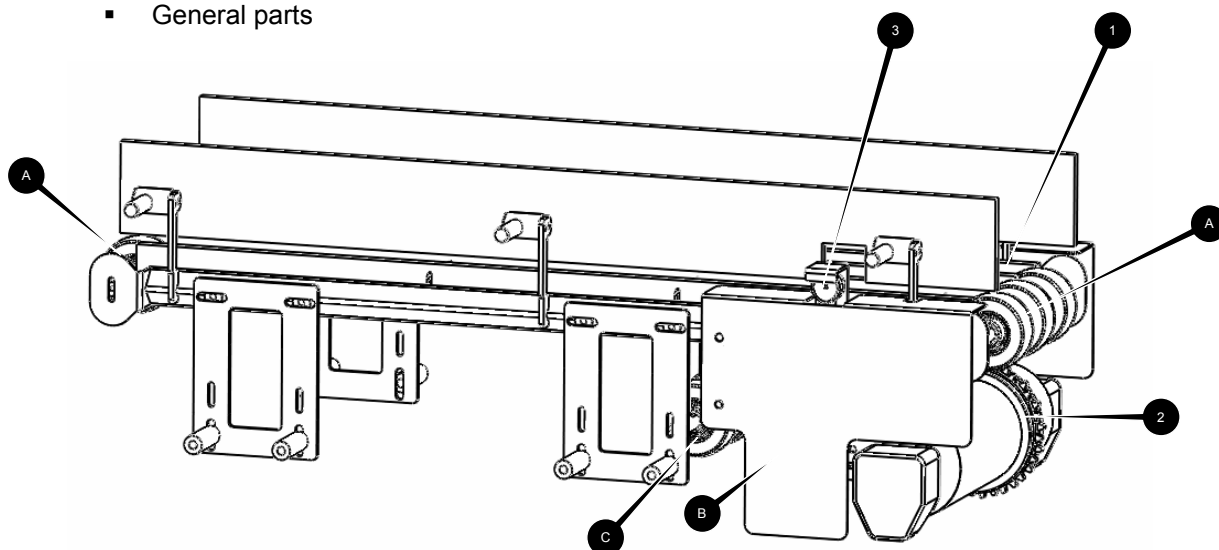
Similar to shaft A

- Shaft C

Similar to shaft A & B

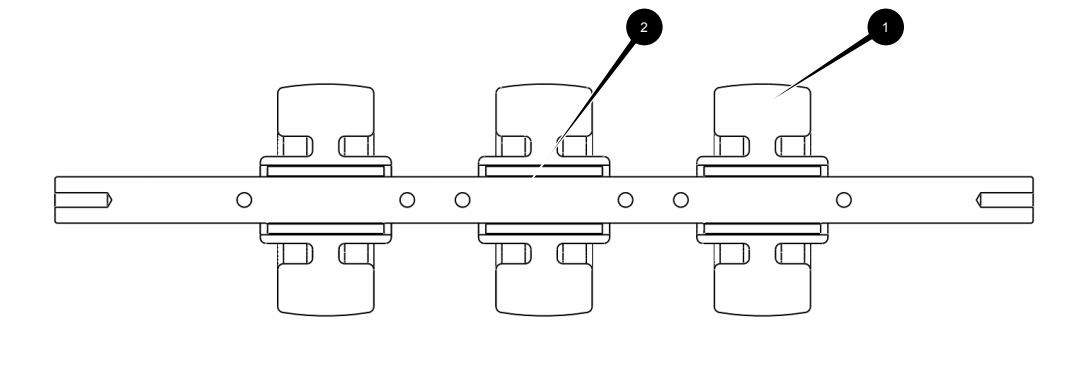
8.4 Second acceleration belt

General parts



Item	Description	Identity
	Belt M1220 High grip	G1568
	Drummotor	G1567
	Sprocket M1236	A312
1	Wearstrip	02/C852
2	Inductive sensor	G485
3	Detection eye receiver	G1234
3	Detection eye sender	G1233
A	Shaft A	
B	Shaft B	
C	Shaft C	

- Shaft A



Item	Description	Identity
1	Wheel D80 mm	52/C751
2	Needle bearing 1624/40	G762

- Shaft B

Similar to shaft A

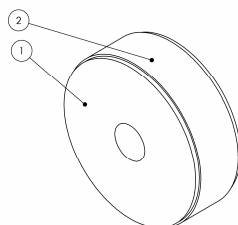
- Shaft C

Similar to shaft A & B

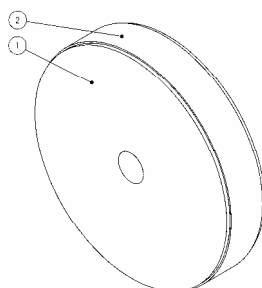
8.5 Grading belt

Item	Description	Identity	Quantity
	Modular Belt Mesh Top M2511 W300mm	G822	1
	Drum motor TM113 L415 mm	G357	1
	Sprocket M25 S18 for drummotor TM113	A513	4
	Proximity sensor M30	G485	1

■ Belt wheels

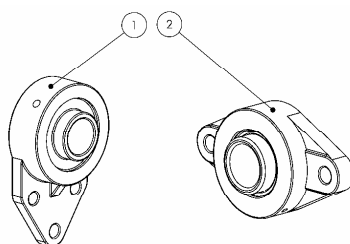


Item	Description	Identity	Quantity
1	Belt wheel Ø90xØ20	24/C464	12
2	Rubber for belt wheel	G129	12



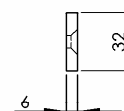
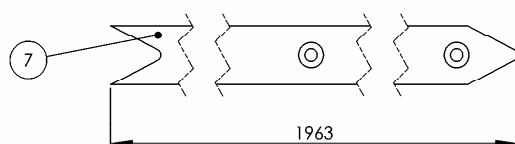
Item	Description	Identity	Quantity
1	Belt wheel Ø150xØ20	25/C464	18
2	Rubber for belt wheel	G130	18

- Bearings



Item	Description	Identity	Quantity
1	3 points bearing Ø20	G363	6
2	2 points bearing Ø20	G364	12

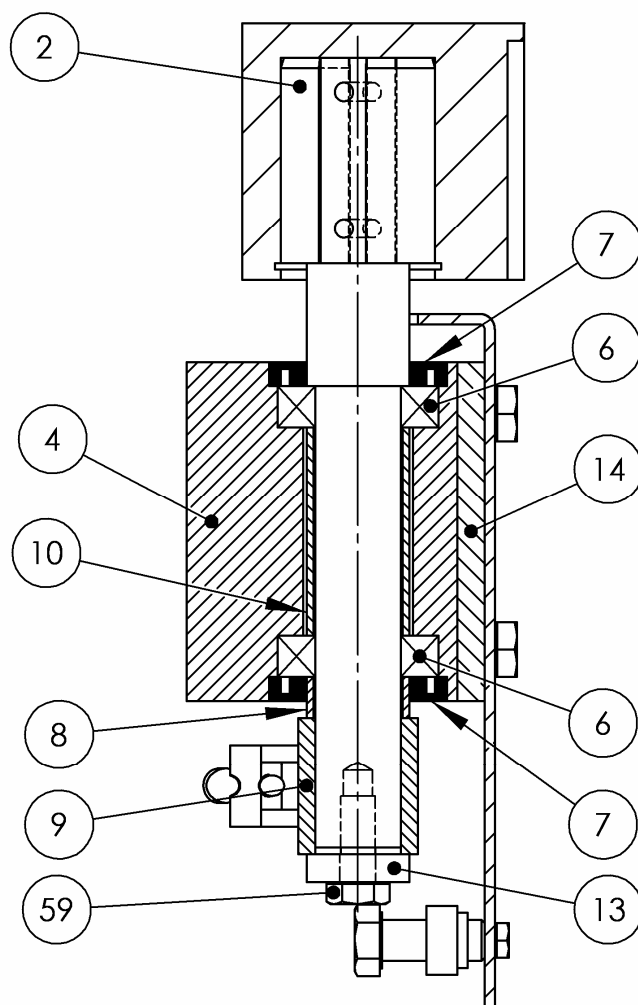
- Wearstrips

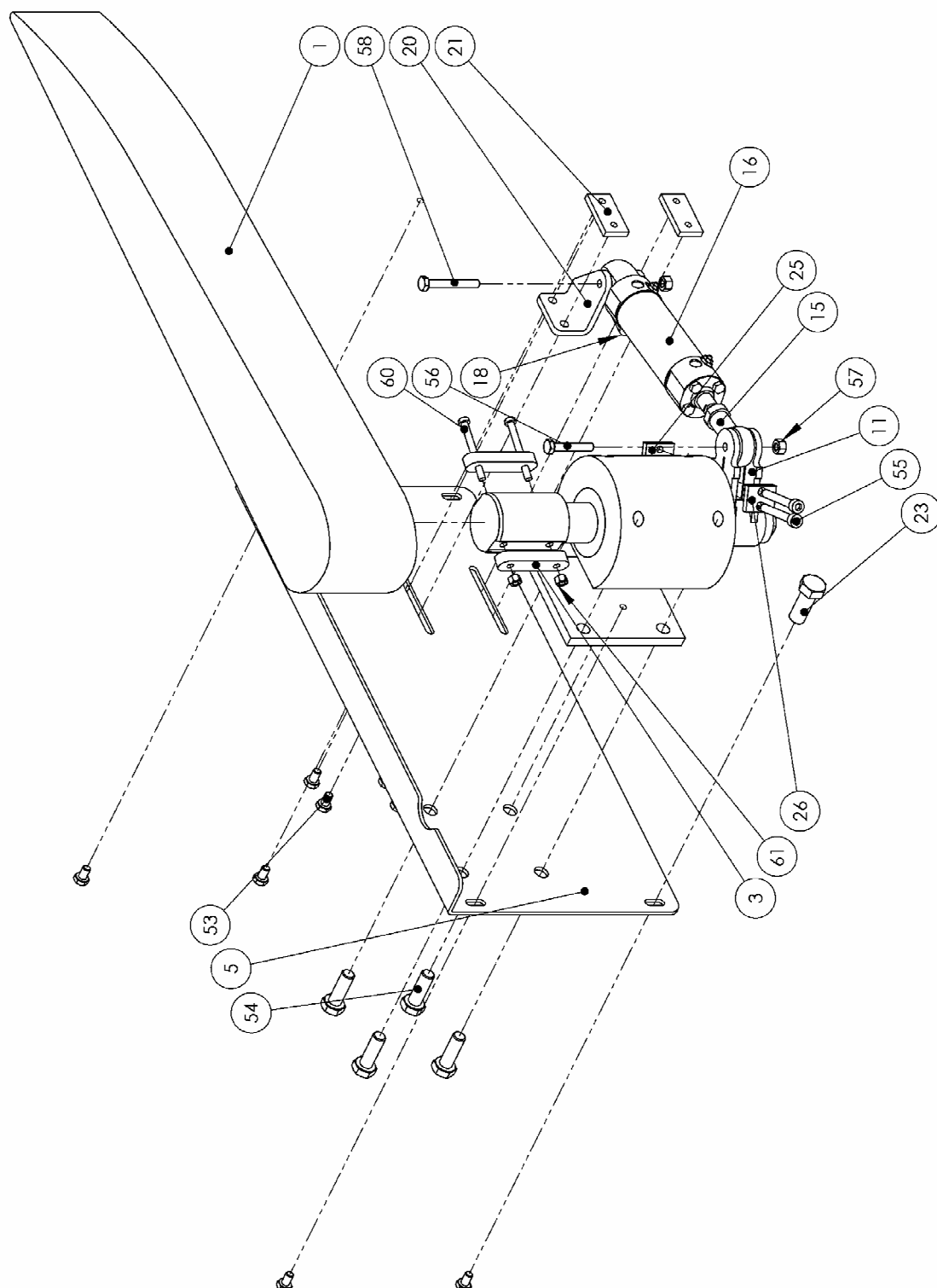


Item	Description	Identity	Quantity
7	Top wearstrip	09/C575	20

Remember that one wearstrip 09/C575 can be replaced by 4 wearstrips with ID G128.

- Flipper assembly





This drawing is the mirrored version of the flipper assembly that is mounted on board of vessel Tonsnes.

Item	Description	Identity	Qty. / flipper module
1	Flipper (mirror)	08A/C395	1
2	Spindle	01/C471	1
3	Key	29/C389	2
4	Bearing house	02/C471	1
5	Flipper frame	29A/C479	1
6	Bearing	6005 2RSR	2
7	Seal	AS 30x52x7	2
8	Tube	06/C471	1
9	Tube	03/C471	1
10	Tube	05/C471	1
11	Cylinder arm	12A/C487	2
12	Tube ballbearing	45A/C332	1
13	Washer	08/C471	1
14	Distance plate	09/C471	1
15	Ballbearing	kbrm10-F	1
16	Cylinder Ø32 L50	G362	1
18	Cylinder foot	09/C395	1
19	Tube cylinder	10/C395	1
20	Mirror cylinder foot	09A/C395	1
21	Nut plate	31/C389	2
23	Bolt M12-M6	17/C464	1
25	Nut plate	72/C549	1
26	Nut plate	71/C549	1
53	Bolt	M6	4

54	bolt	M10	4
55	bolt	DIN 912 M6	2
56	bolt	M6	1
57	nut	M6	2
58	bolt	M6	1
59	bolt	M10	1
60	Bolt	DIN 84 M5	2
61	Nut	M5	2